

BFS Traversal

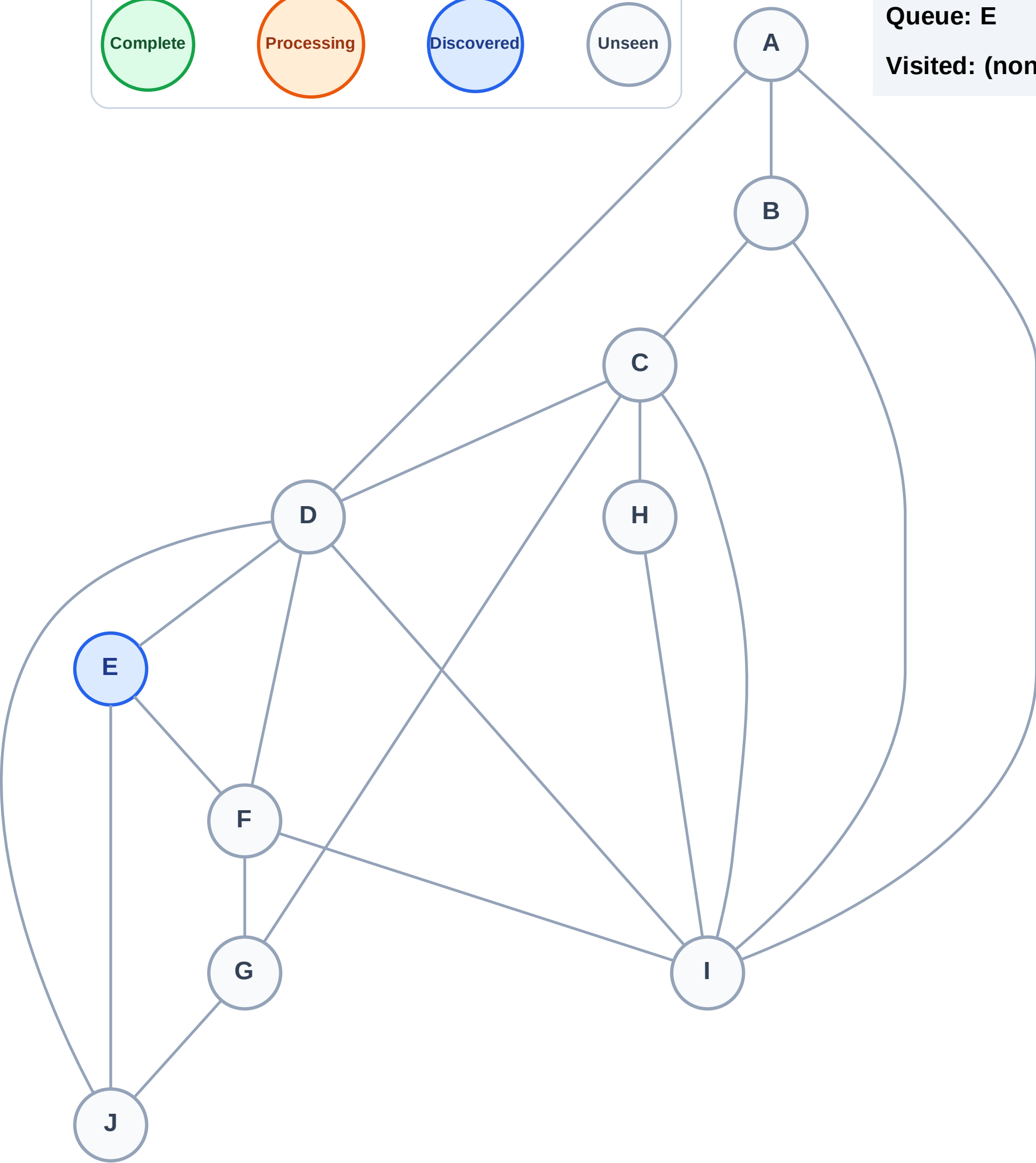
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

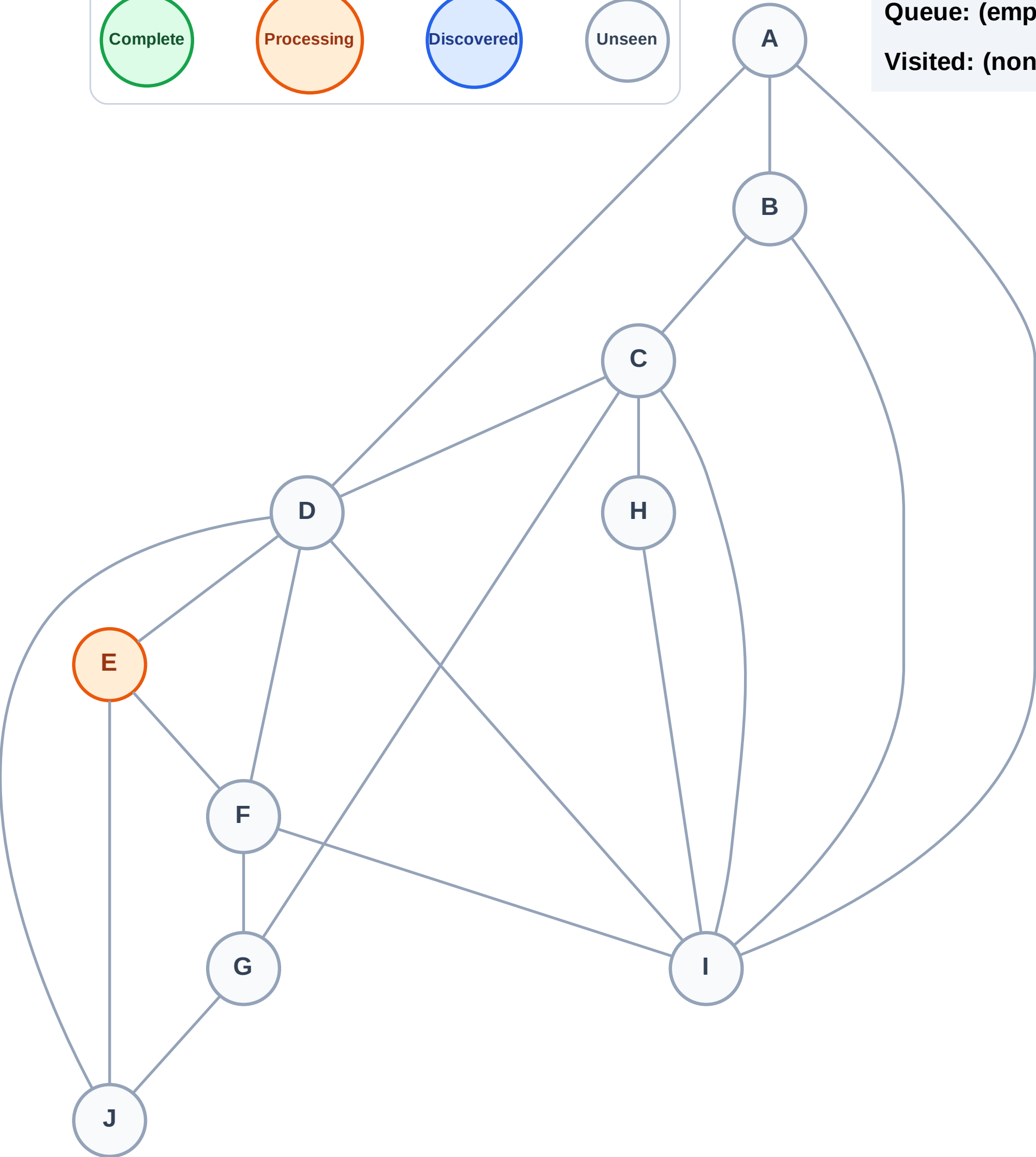
Complete

Processing

Discovered

Unseen

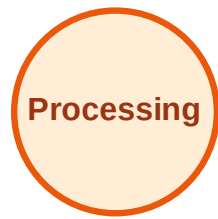
Queue: (empty)
Visited: (none)



BFS Traversal

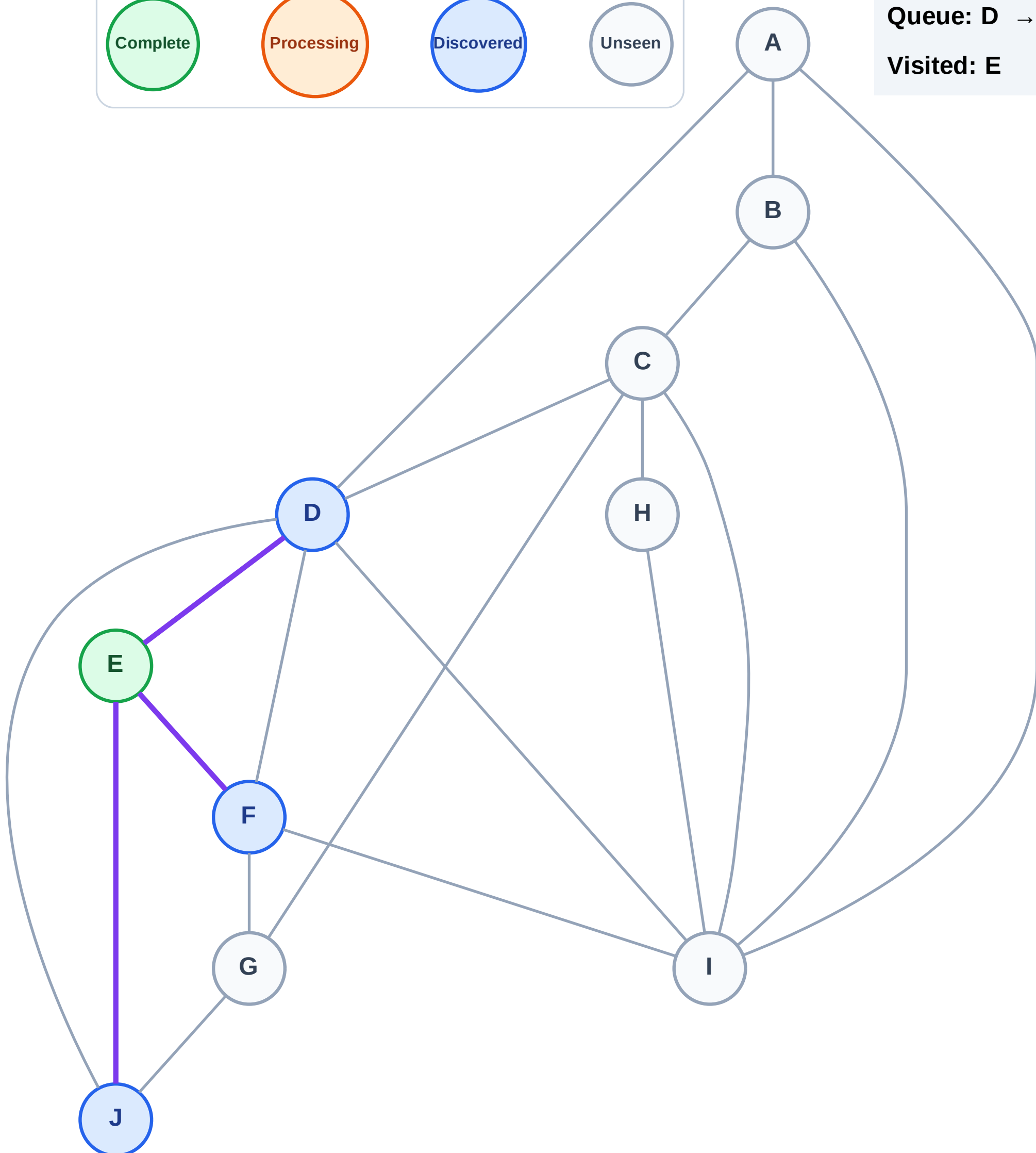
Step 3: Discover D, F, J from E

Legend



Queue: D → F → J

Visited: E



BFS Traversal

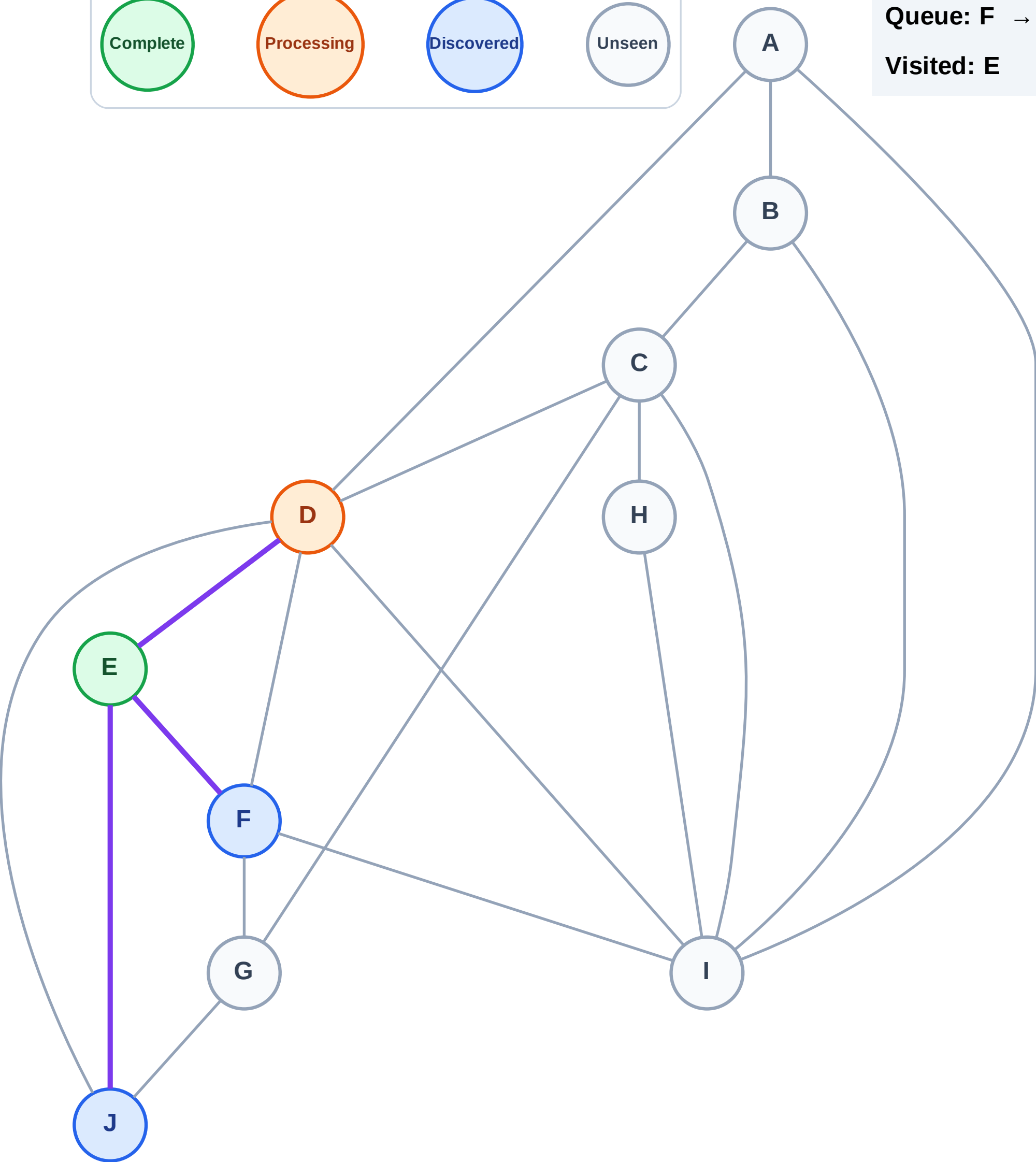
Step 4: Process node D

Legend



Queue: F → J

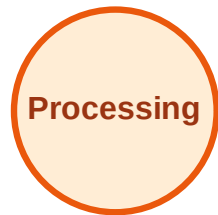
Visited: E



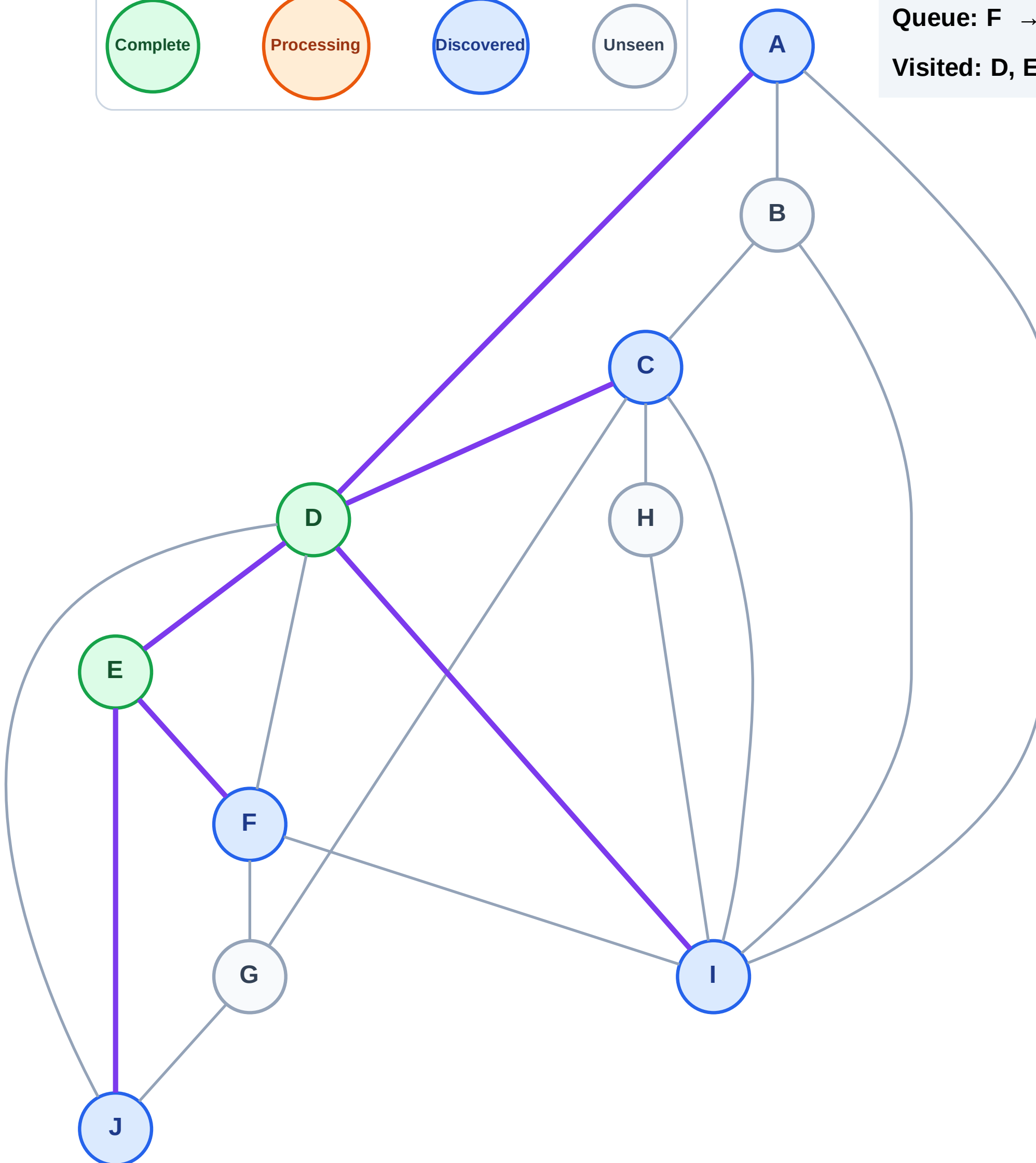
BFS Traversal

Step 5: Discover A, C, I from D

Legend



Queue: F → J → A → C → I
Visited: D, E



BFS Traversal

Step 6: Process node F

Legend

Complete

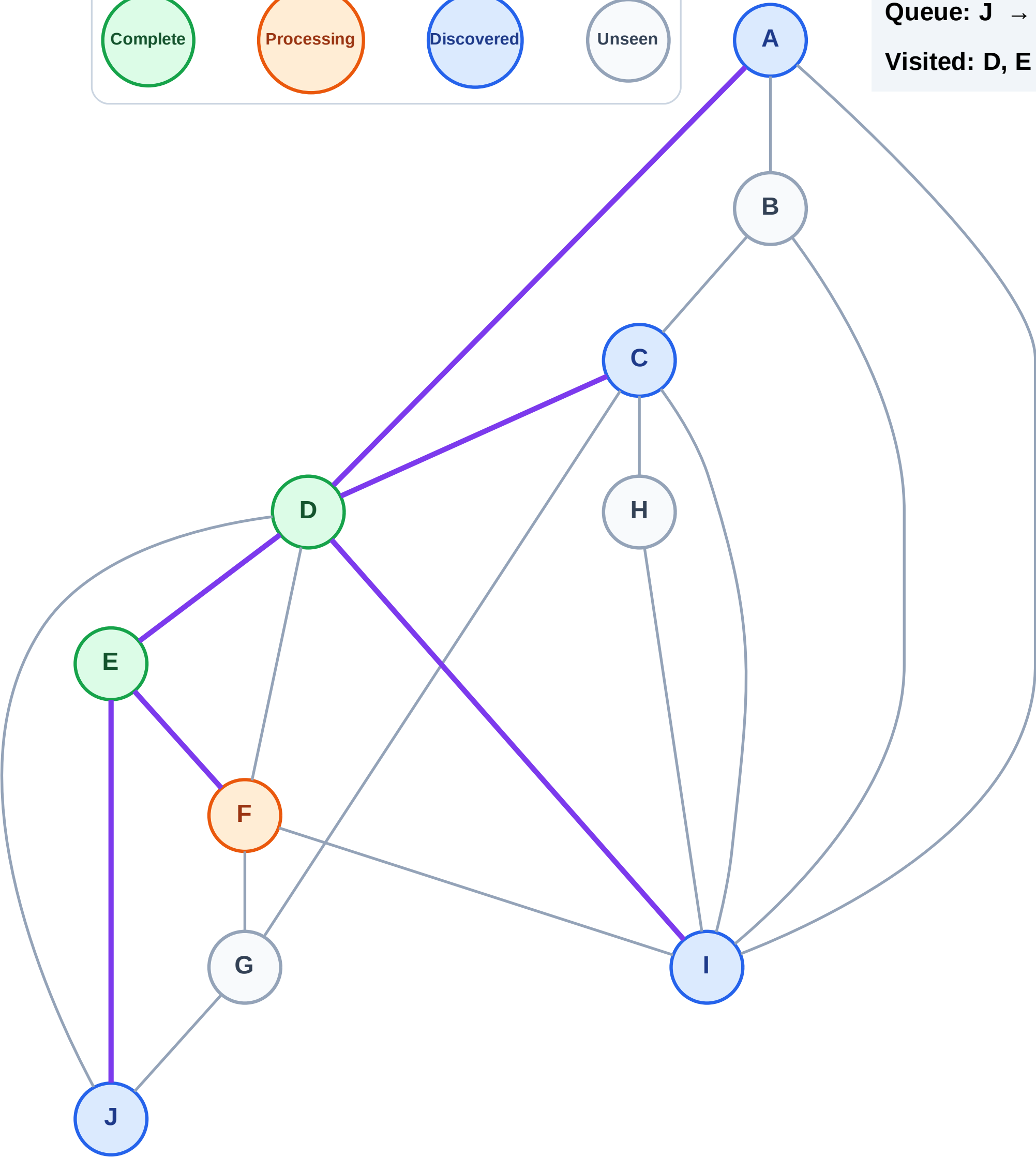
Processing

Discovered

Unseen

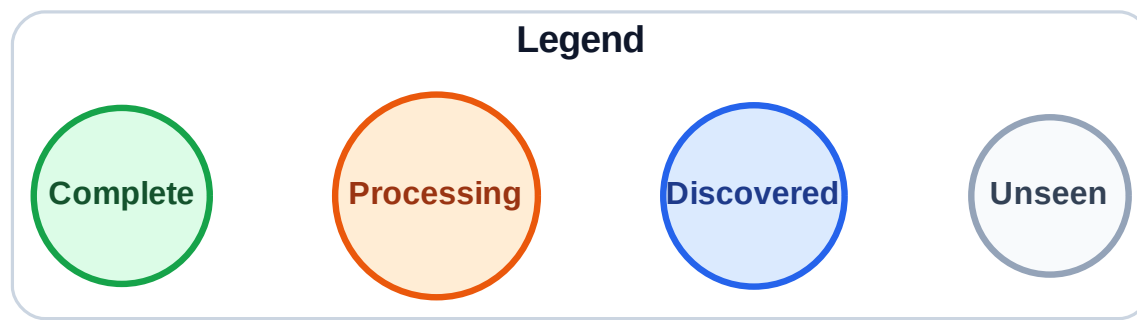
Queue: J → A → C → I

Visited: D, E



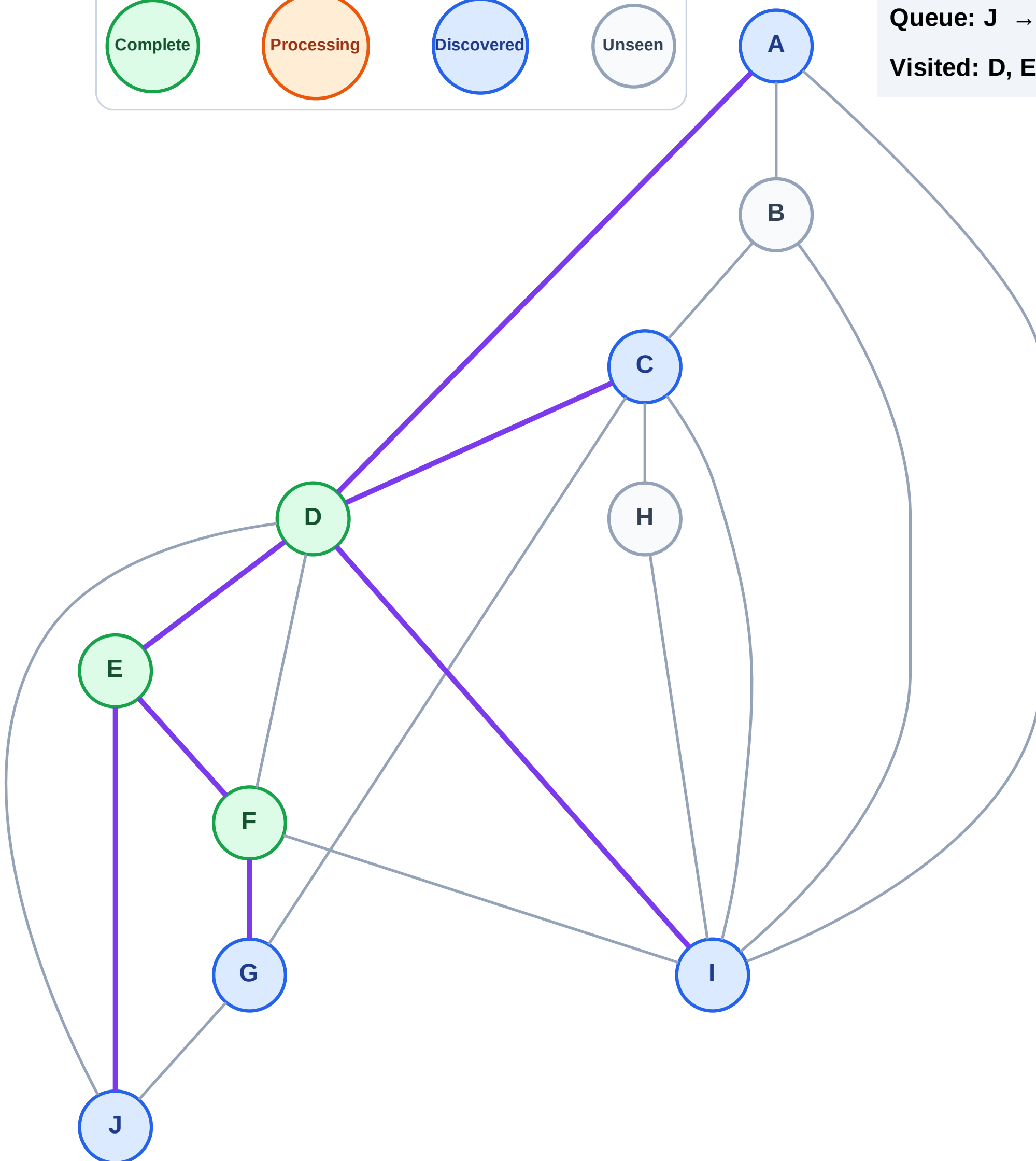
BFS Traversal

Step 7: Discover G from F



Queue: J → A → C → I → G

Visited: D, E, F



BFS Traversal

Step 8: Process node J

Legend

Complete

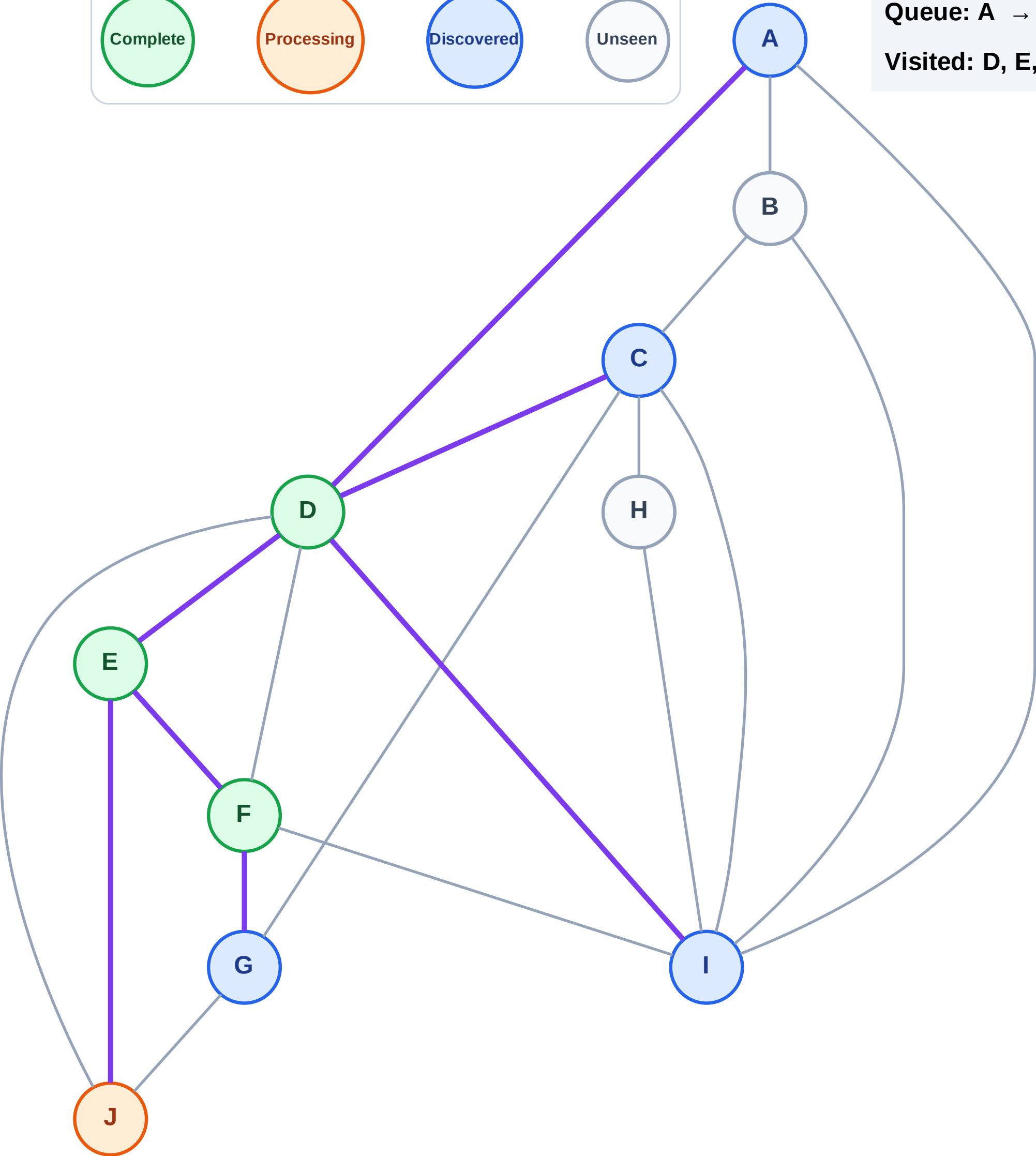
Processing

Discovered

Unseen

Queue: A → C → I → G

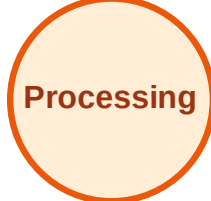
Visited: D, E, F



BFS Traversal

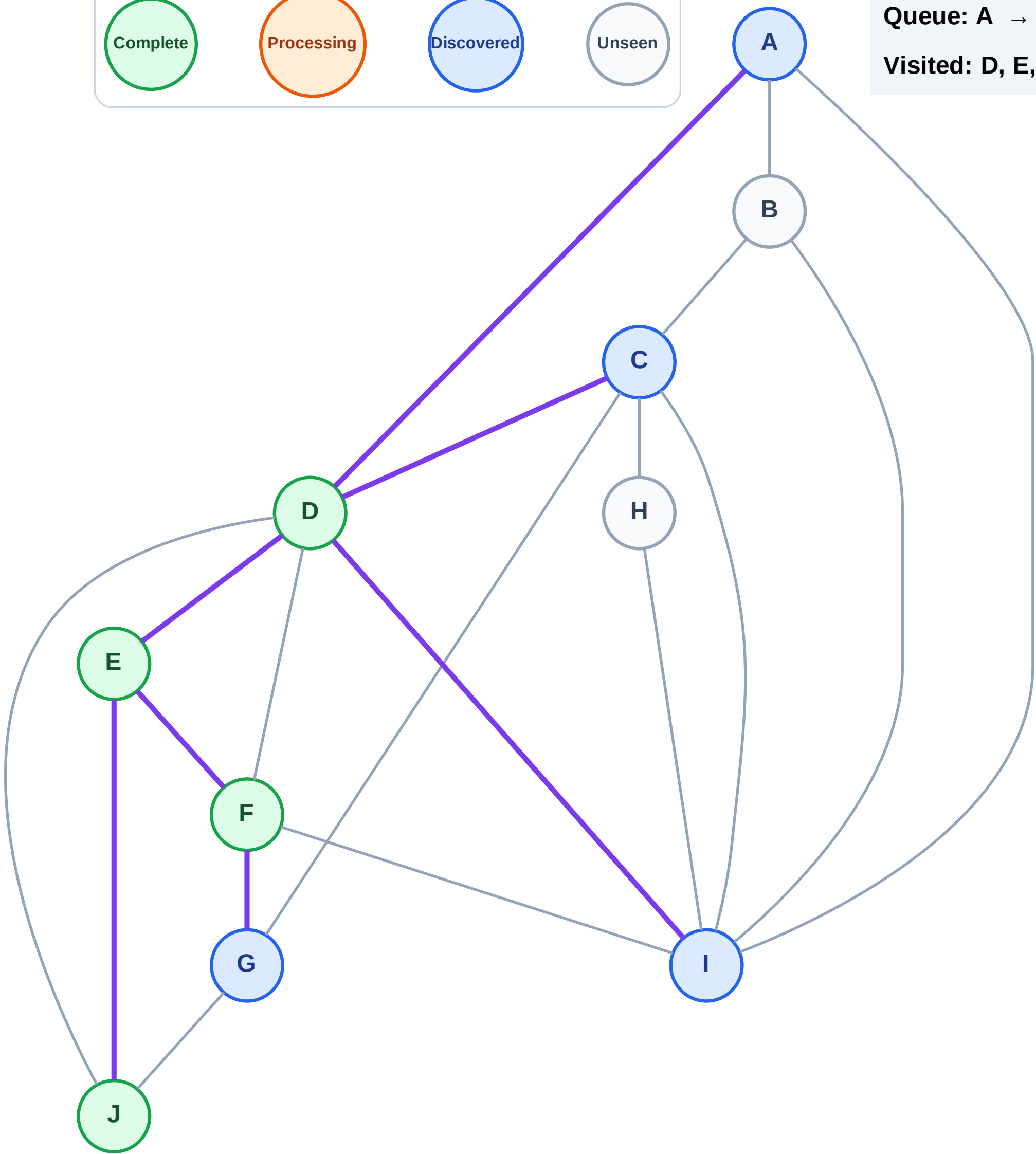
Step 9: No new neighbors from J

Legend



Queue: A → C → I → G

Visited: D, E, F, J



BFS Traversal

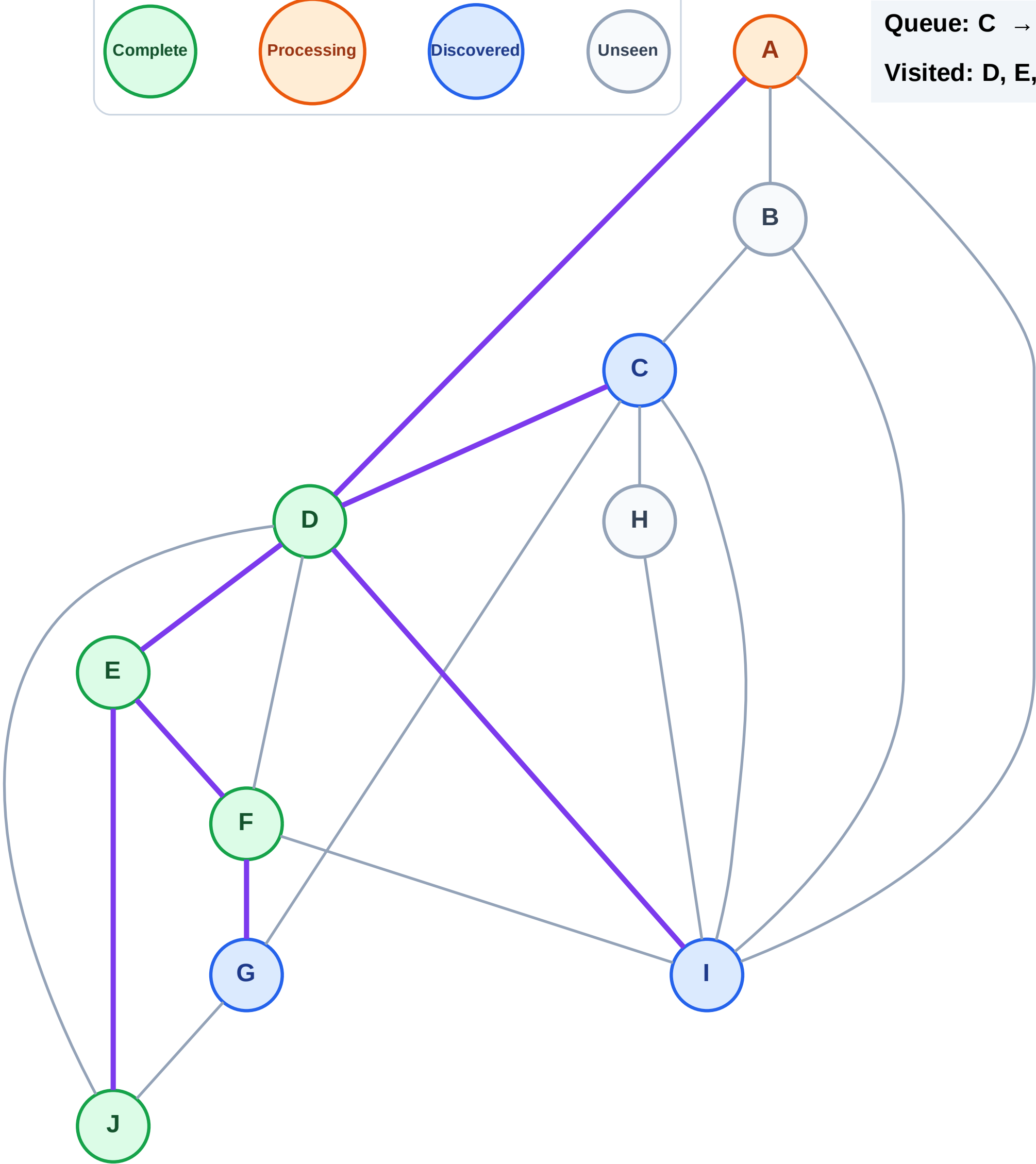
Step 10: Process node A

Legend



Queue: C → I → G

Visited: D, E, F, J



BFS Traversal

Step 11: Discover B from A

Legend

Complete

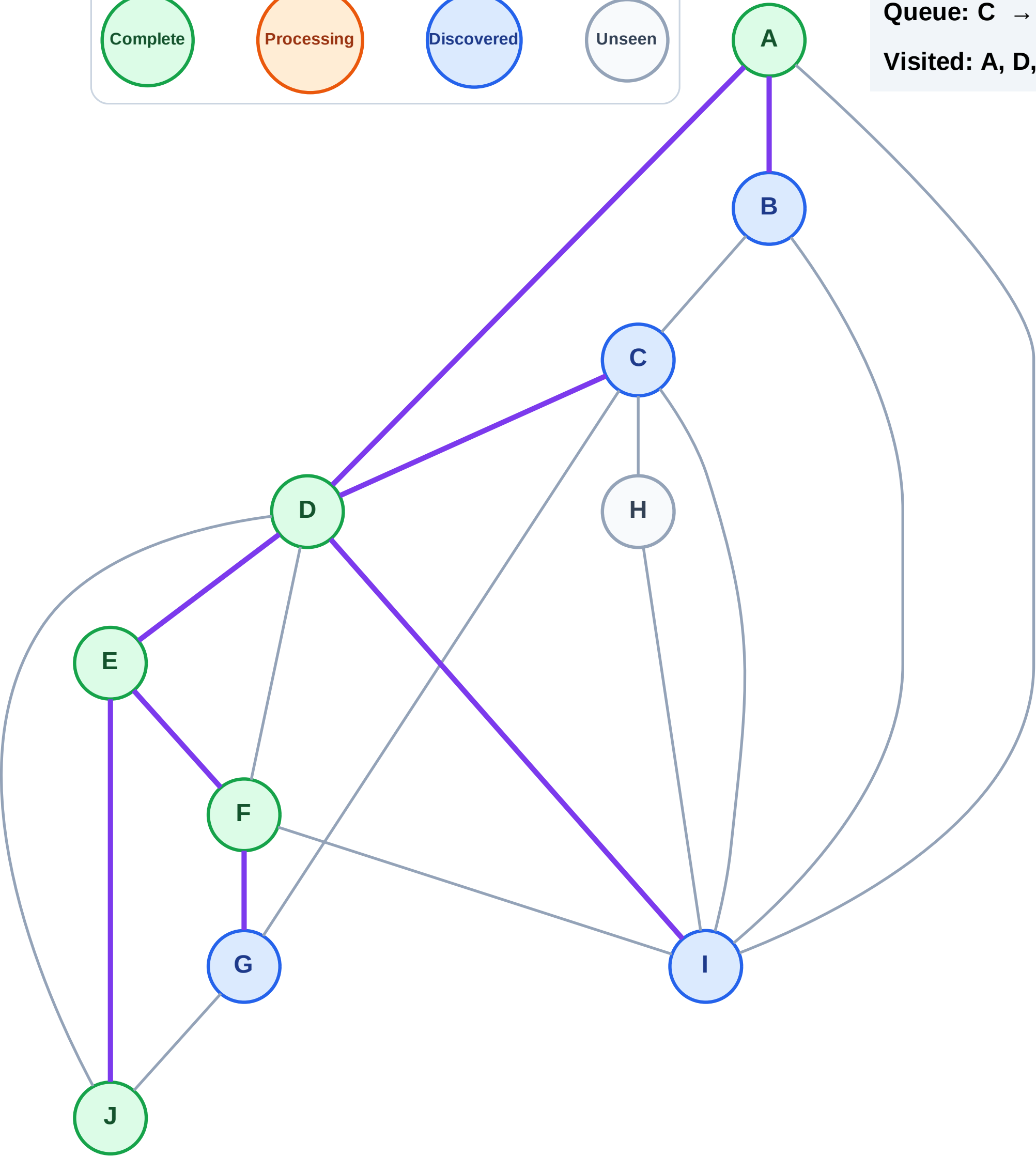
Processing

Discovered

Unseen

Queue: C → I → G → B

Visited: A, D, E, F, J



BFS Traversal

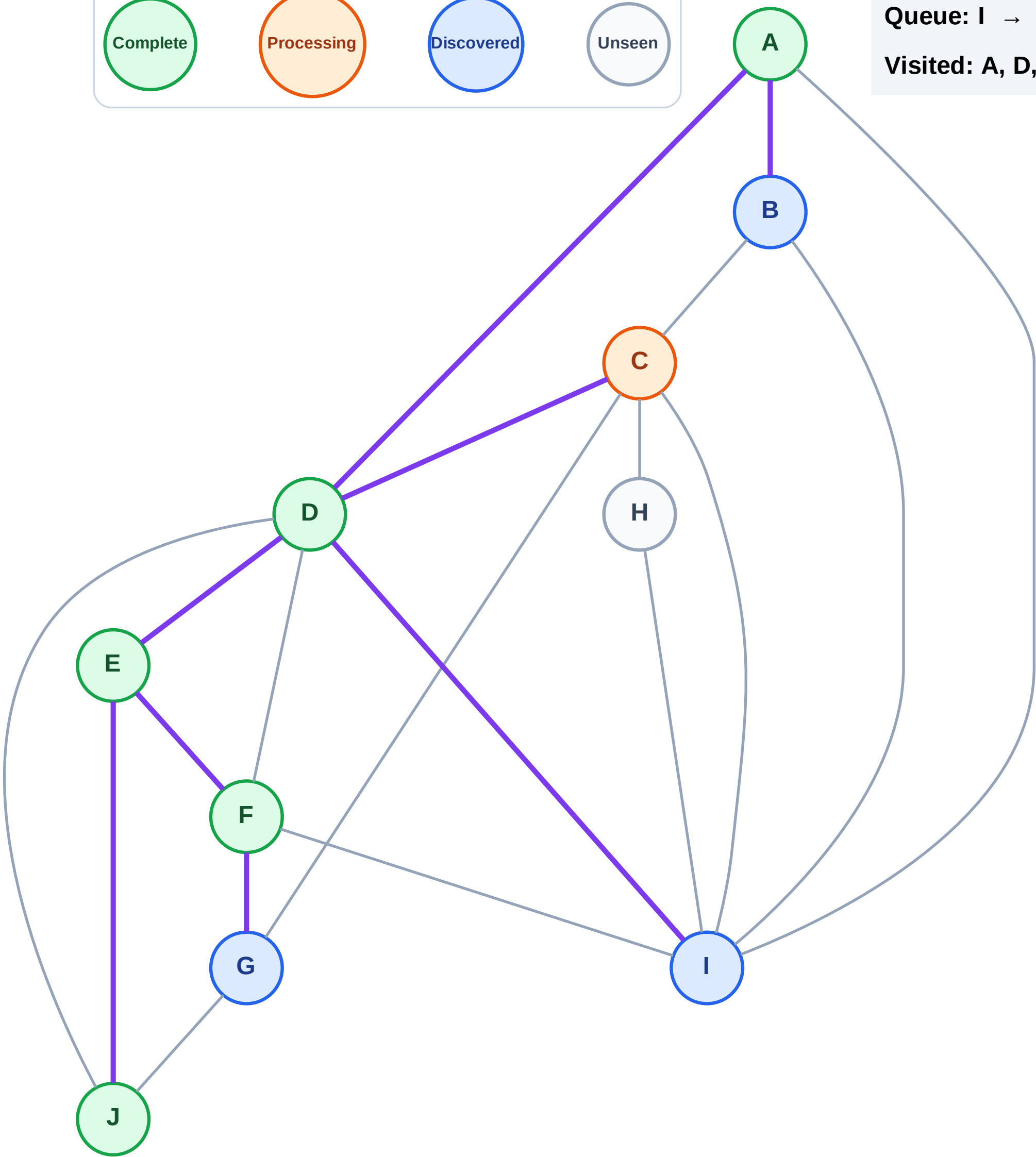
Step 12: Process node C

Legend



Queue: I → G → B

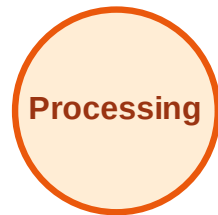
Visited: A, D, E, F, J



BFS Traversal

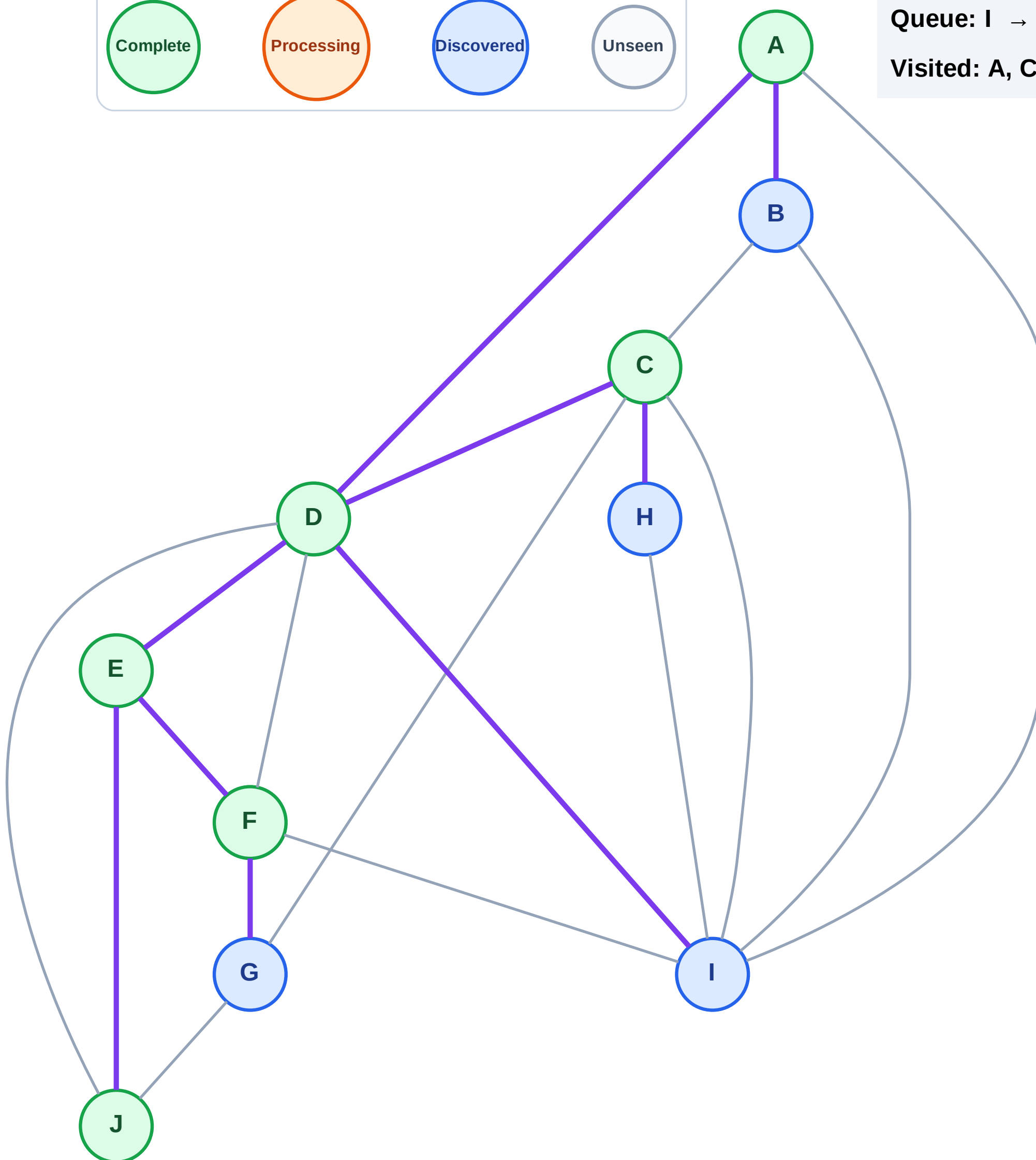
Step 13: Discover H from C

Legend



Queue: I → G → B → H

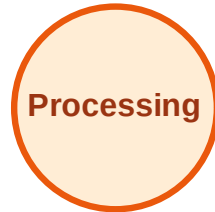
Visited: A, C, D, E, F, J



BFS Traversal

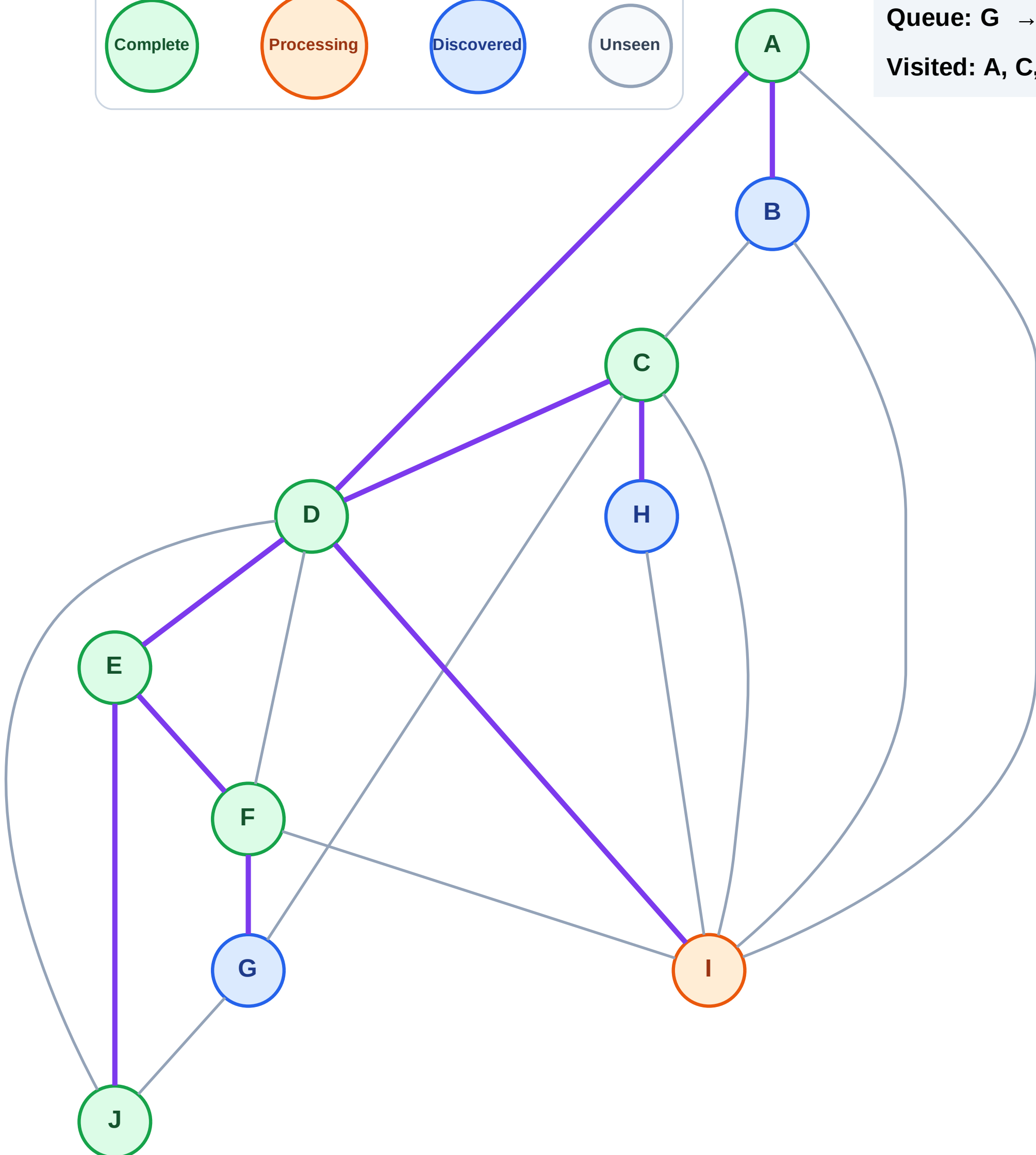
Step 14: Process node I

Legend



Queue: G → B → H

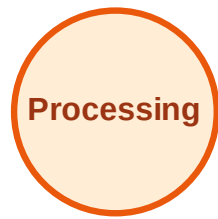
Visited: A, C, D, E, F, J



BFS Traversal

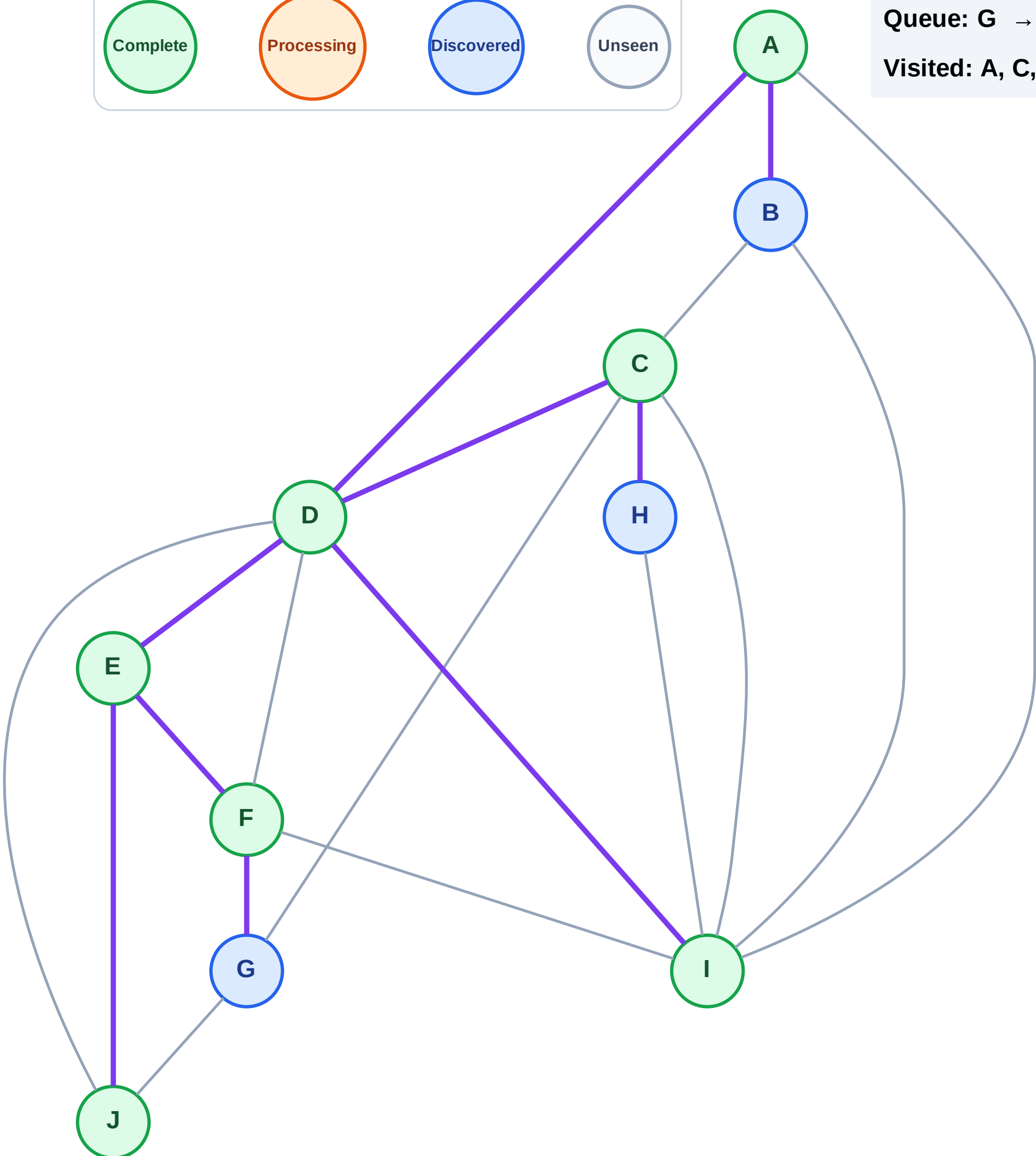
Step 15: No new neighbors from I

Legend



Queue: G → B → H

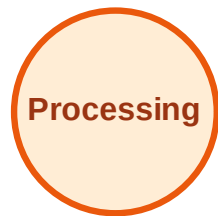
Visited: A, C, D, E, F, I, J



BFS Traversal

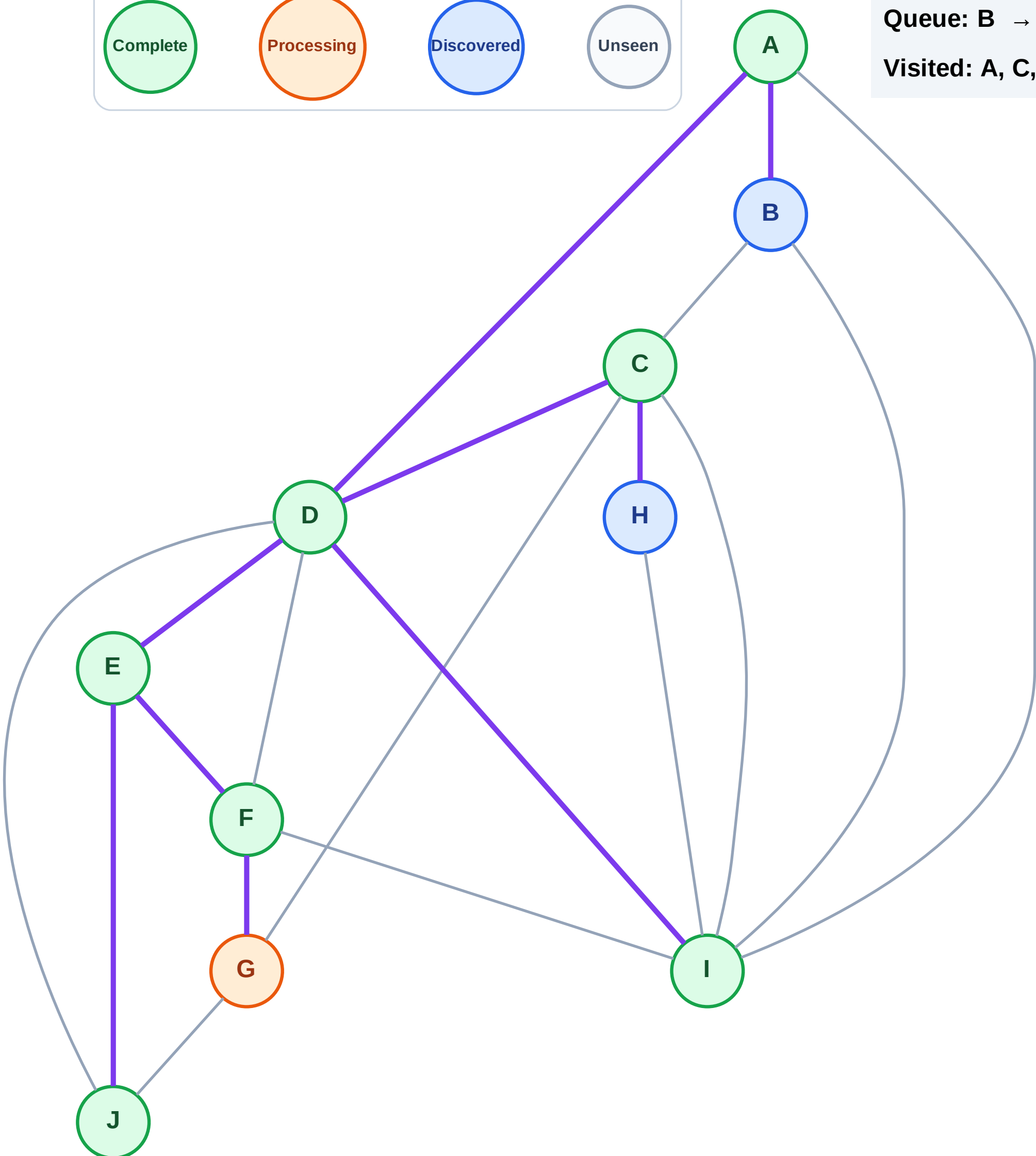
Step 16: Process node G

Legend



Queue: B → H

Visited: A, C, D, E, F, I, J



BFS Traversal

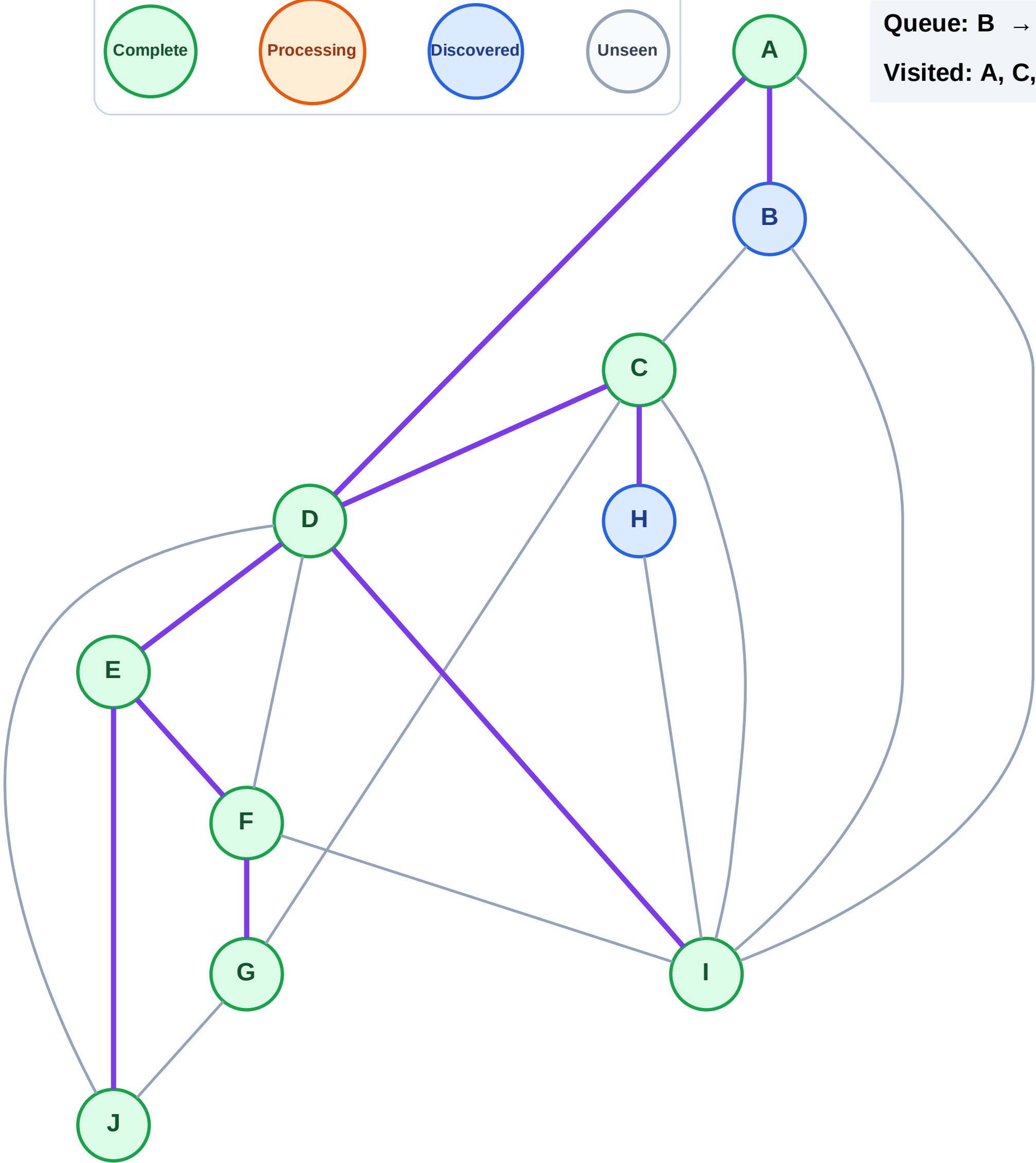
Step 17: No new neighbors from G

Legend



Queue: B → H

Visited: A, C, D, E, F, G, I, J



BFS Traversal

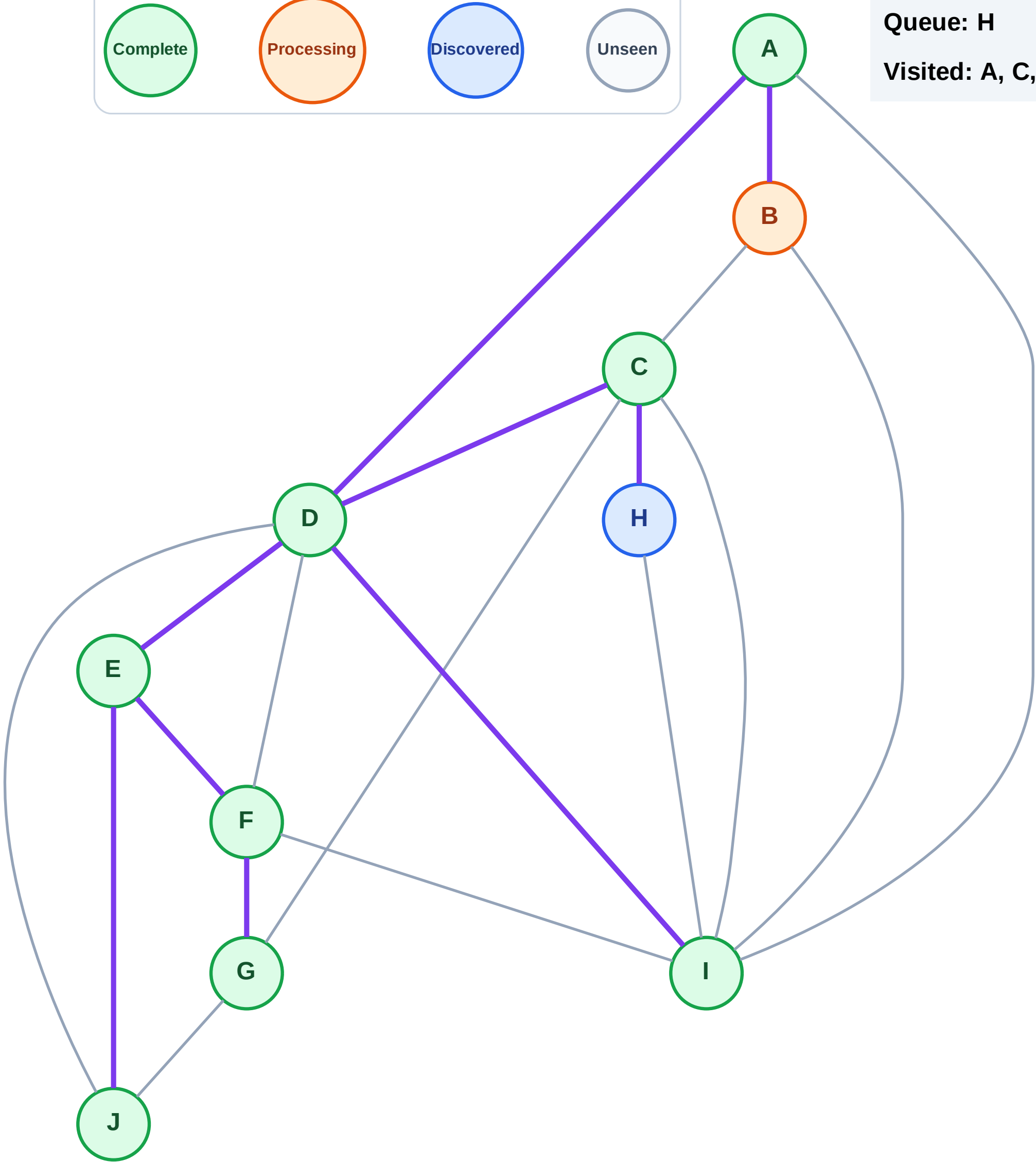
Step 18: Process node B

Legend



Queue: H

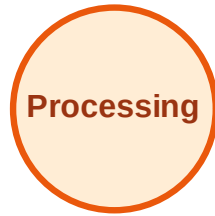
Visited: A, C, D, E, F, G, I, J



BFS Traversal

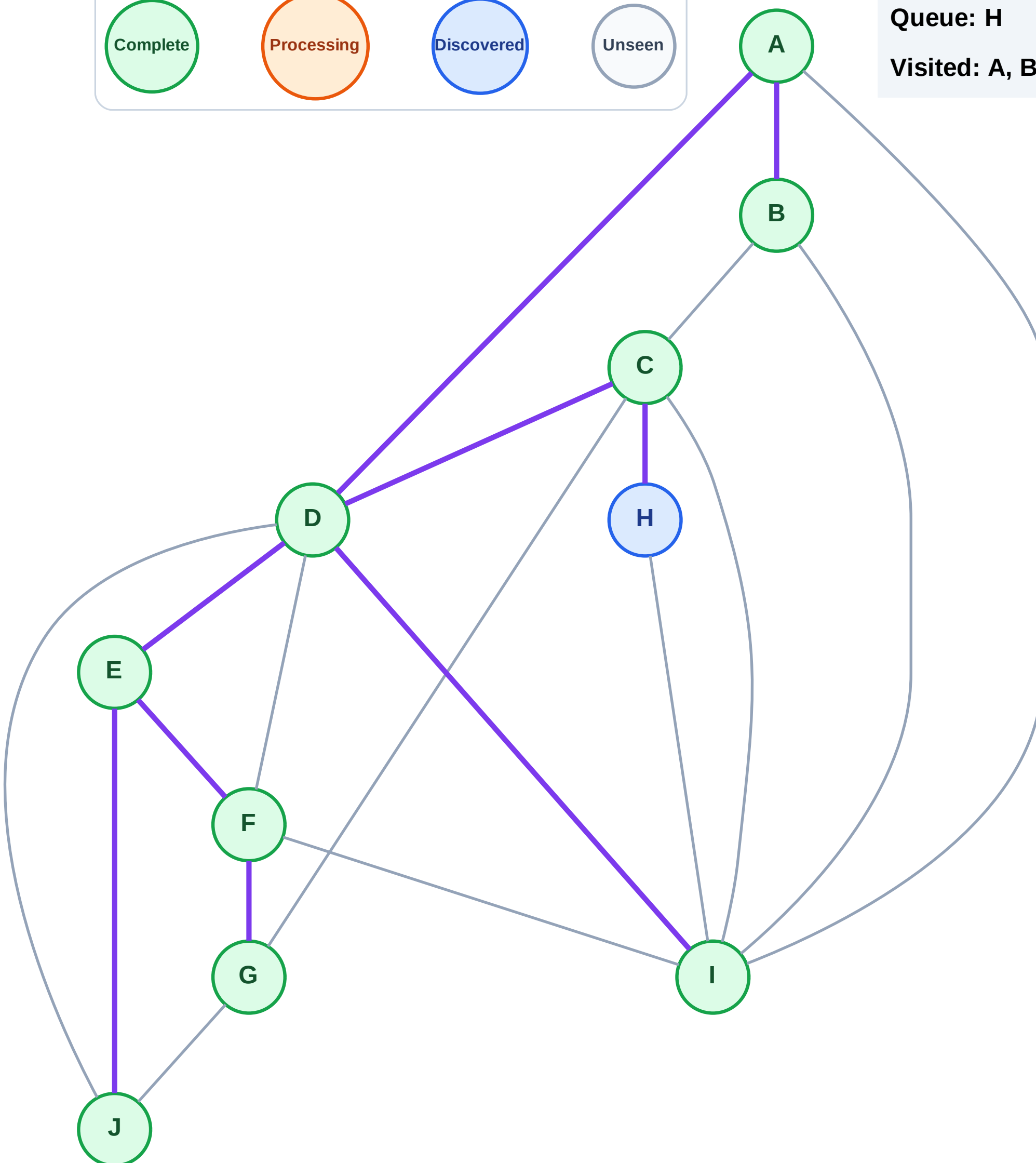
Step 19: No new neighbors from B

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

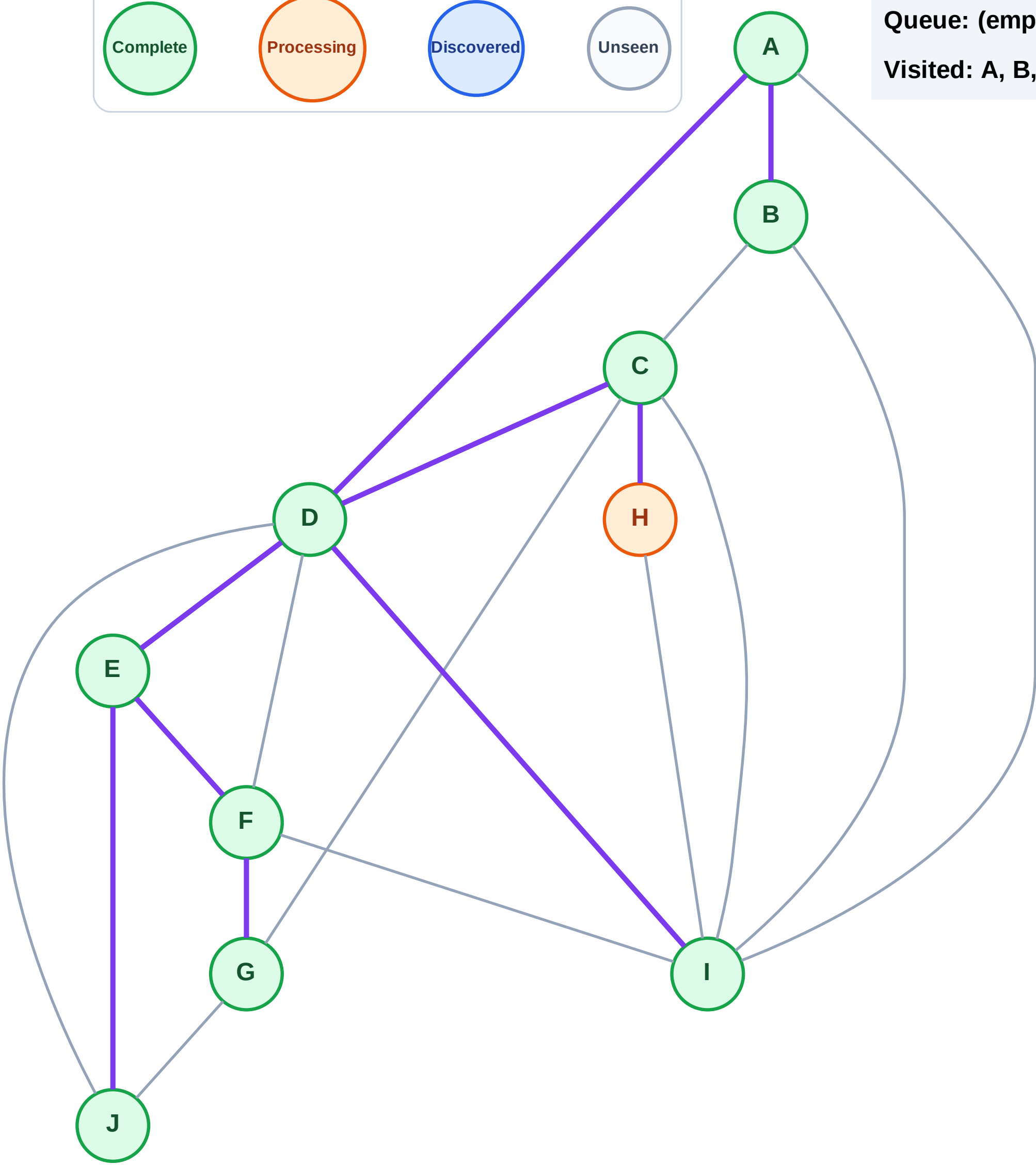
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



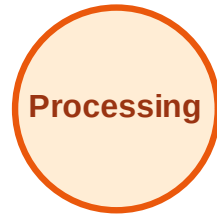
BFS Traversal

Step 21: No new neighbors from H

Legend



Complete



Processing



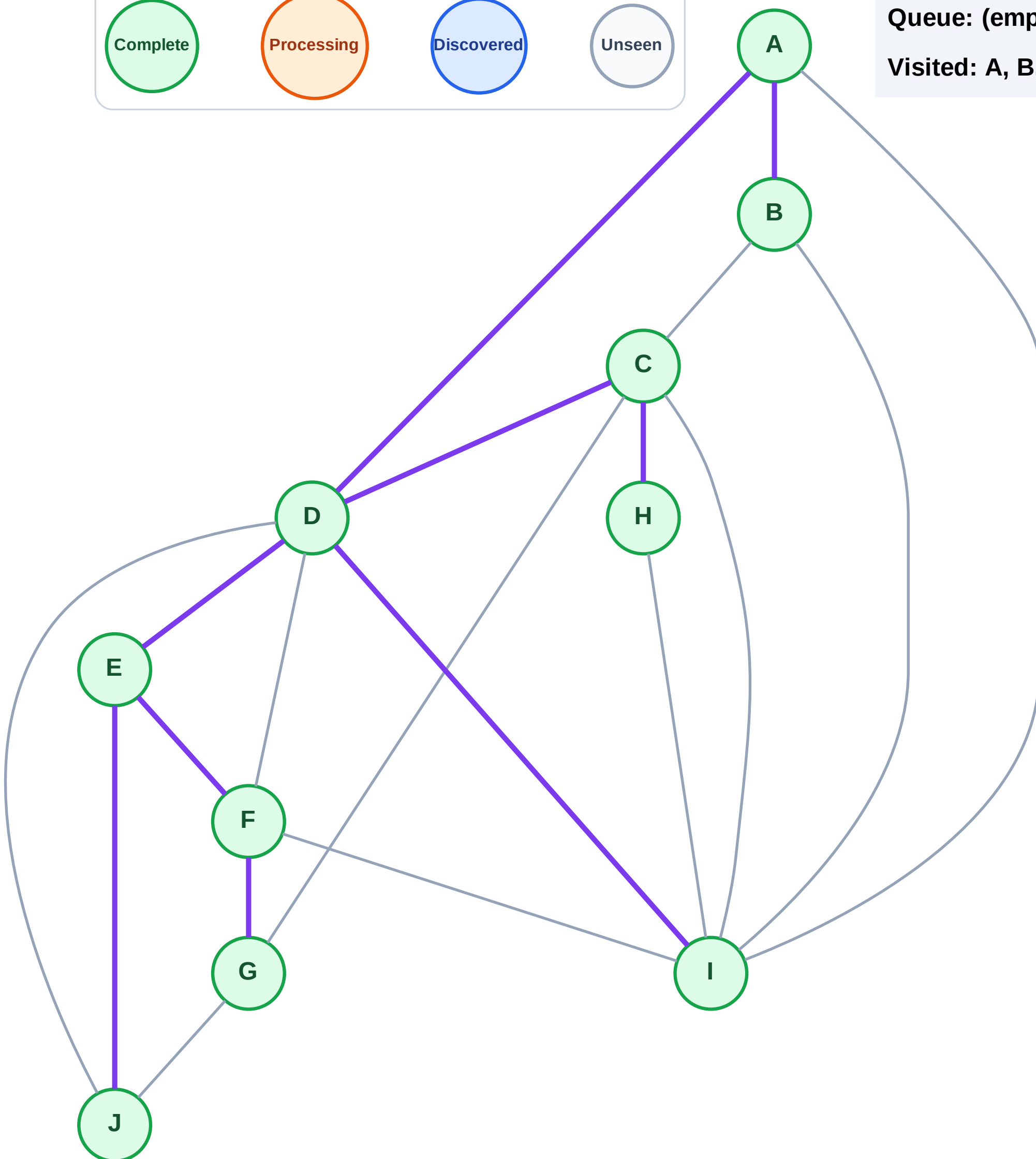
Discovered



Unseen

Queue: (empty)

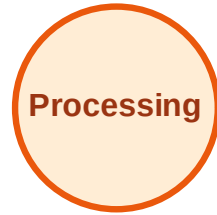
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

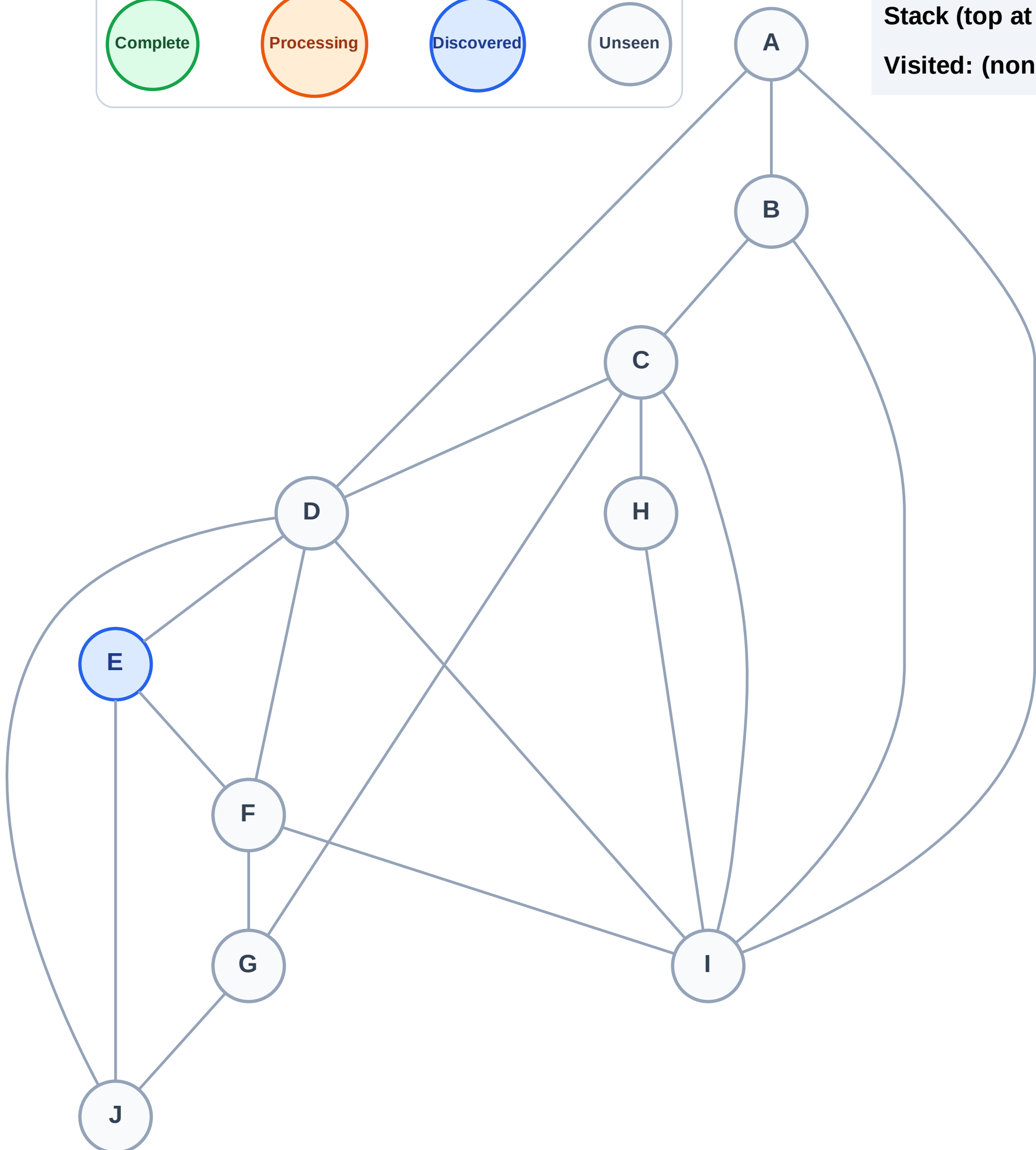
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

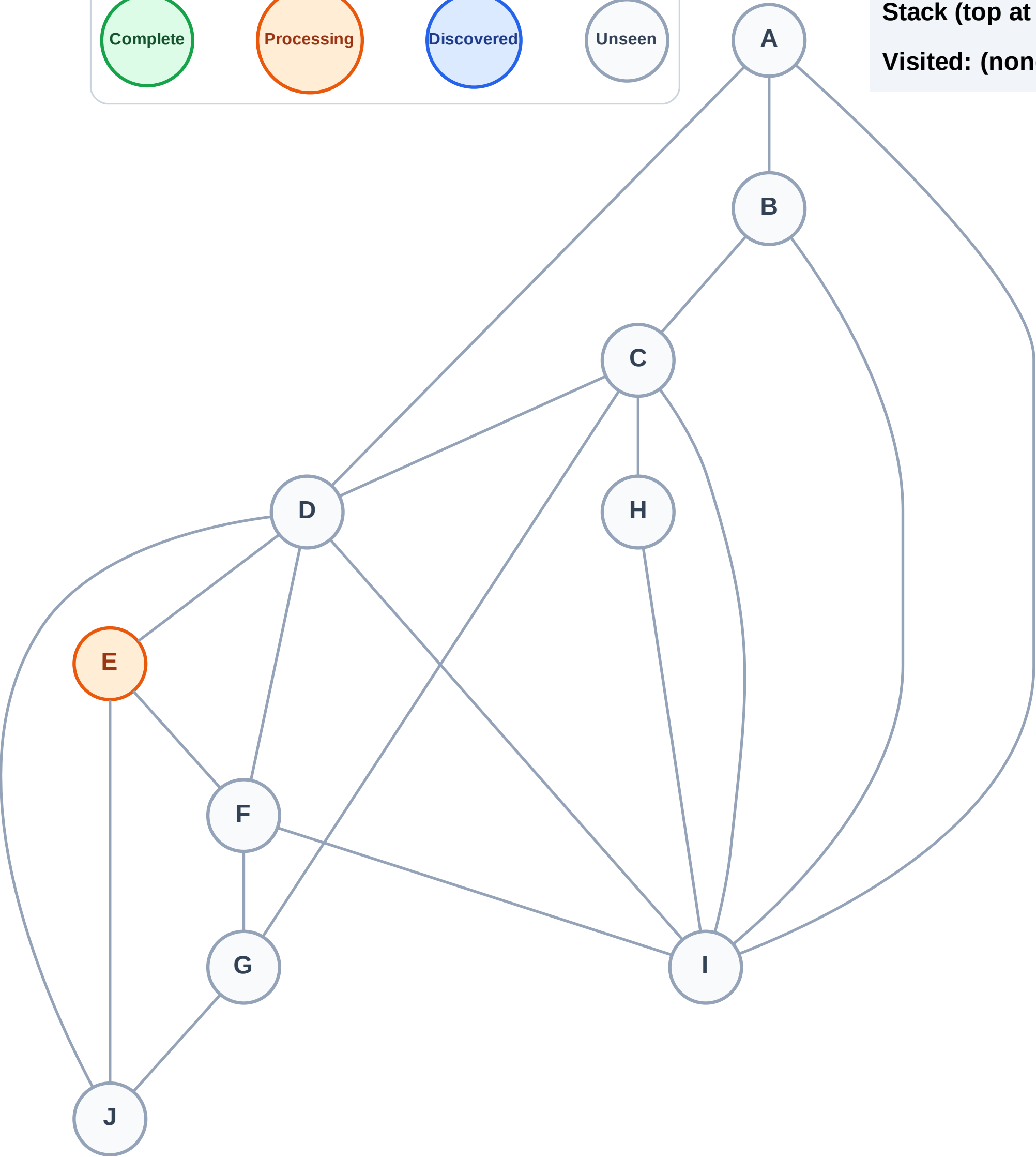
Complete

Processing

Discovered

Unseen

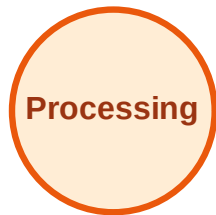
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

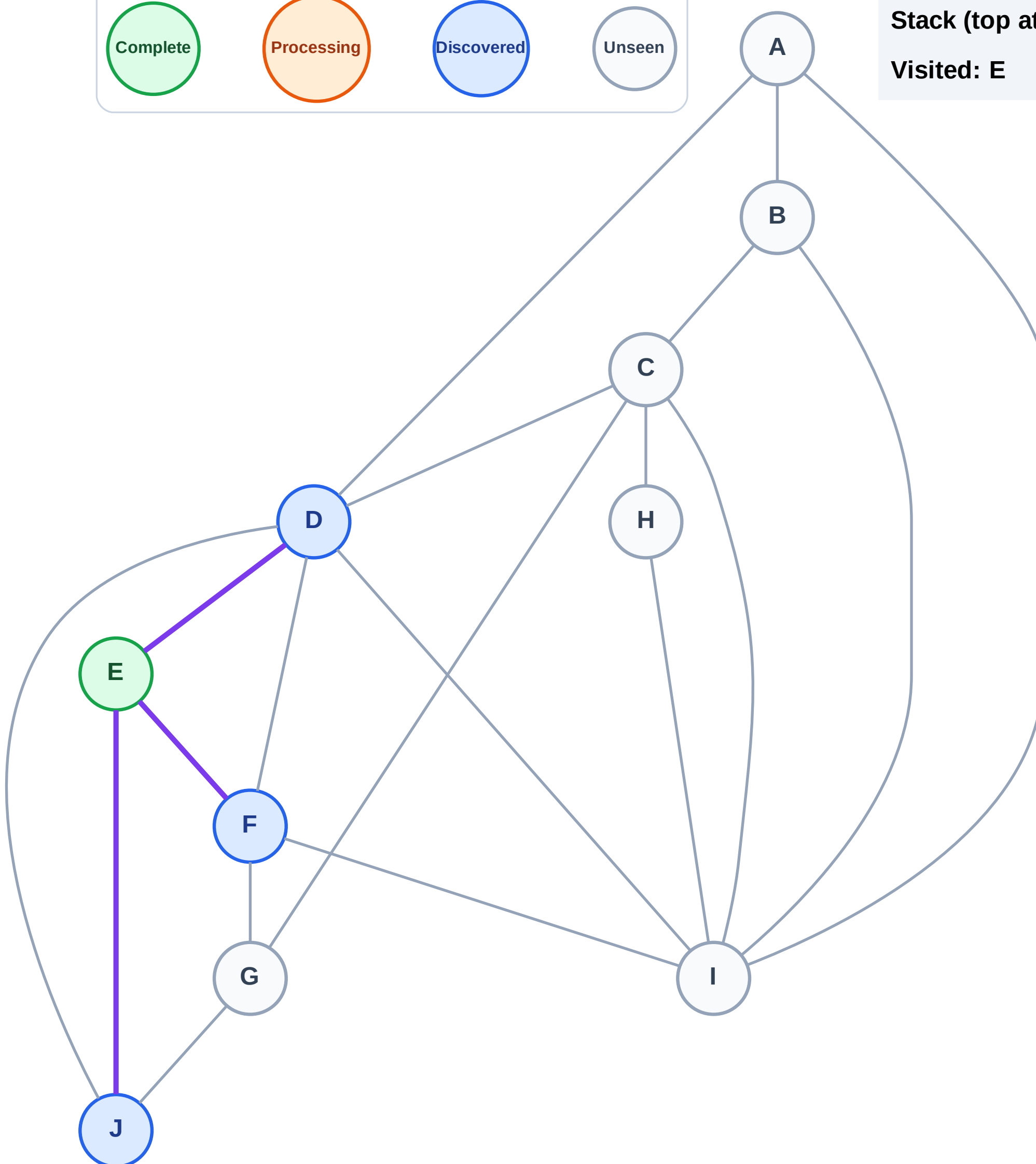
Step 3: Discover J, F, D from E

Legend



Stack (top at right): J | F | D

Visited: E



DFS Traversal

Step 4: Process node D

Legend

Complete

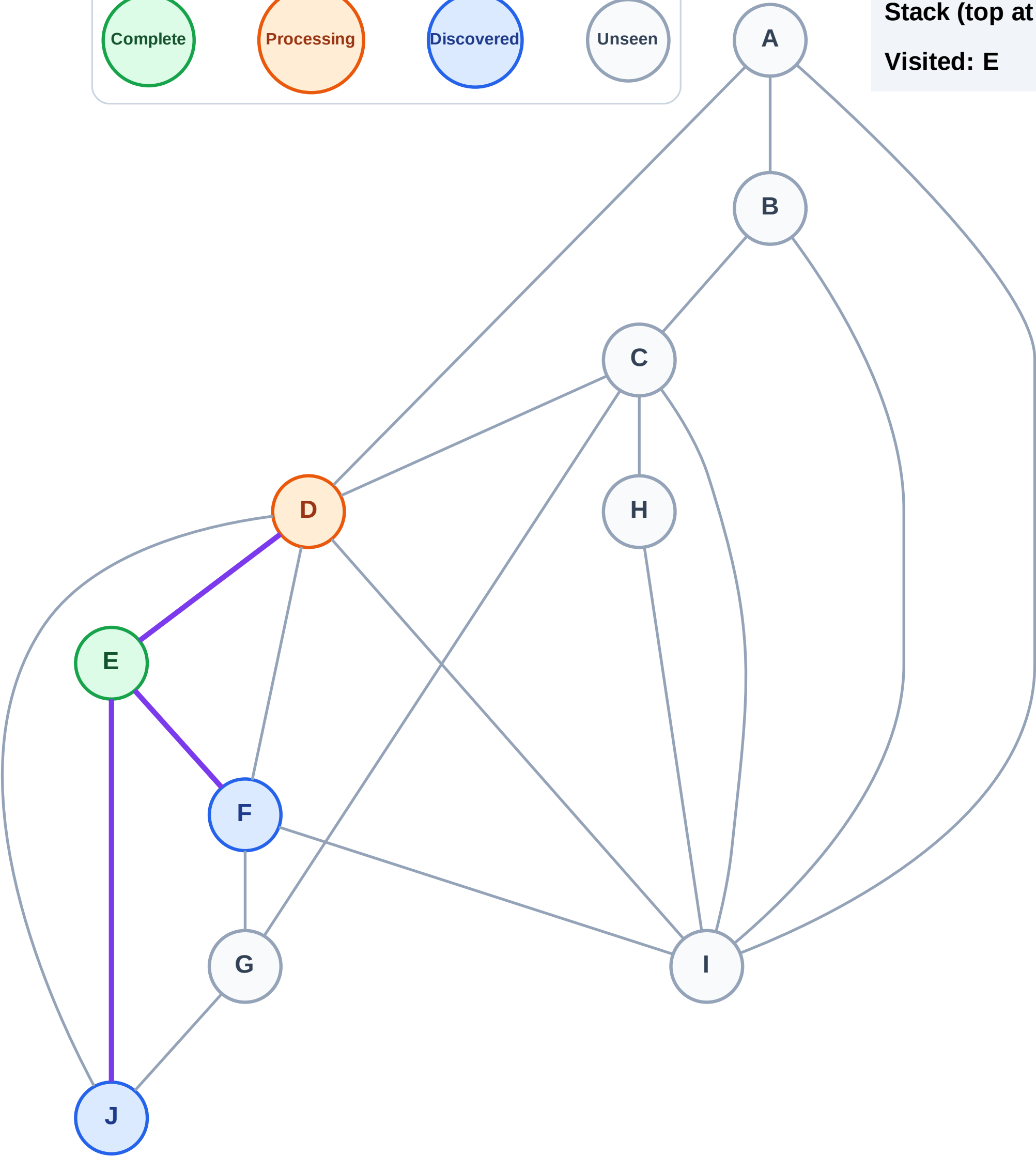
Processing

Discovered

Unseen

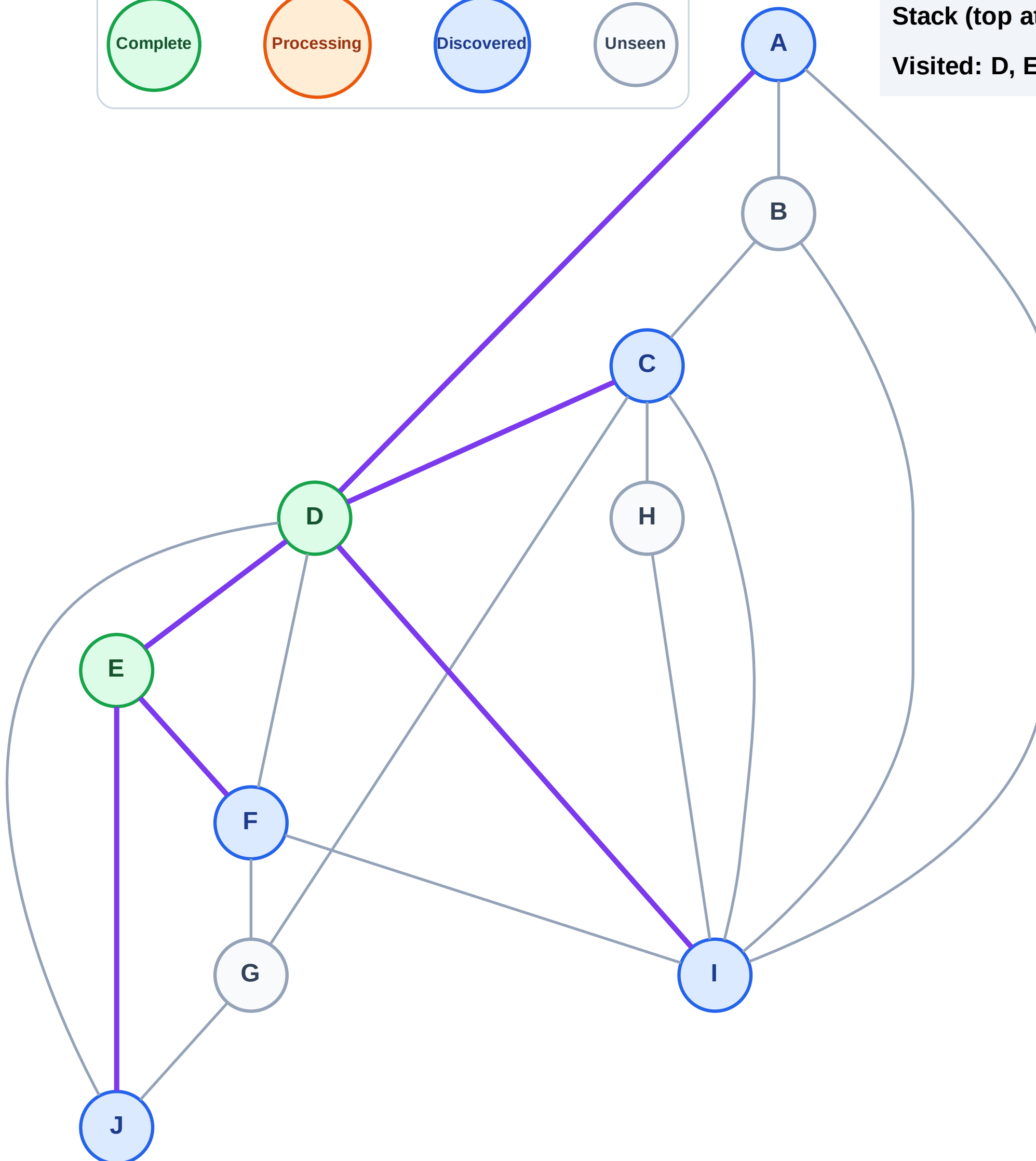
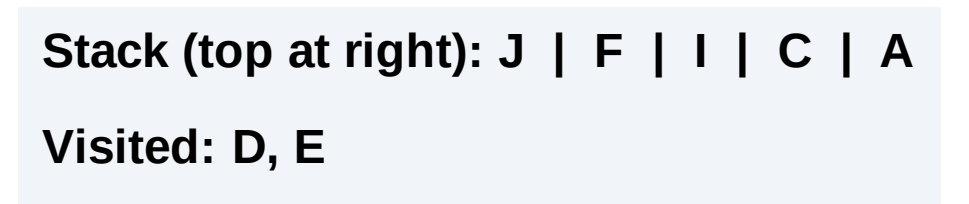
Stack (top at right): J | F

Visited: E



Step 5: Discover I, C, A from D

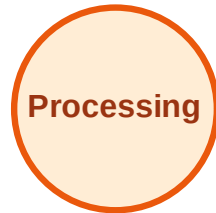
Step 5: Discover I, C, A from D



DFS Traversal

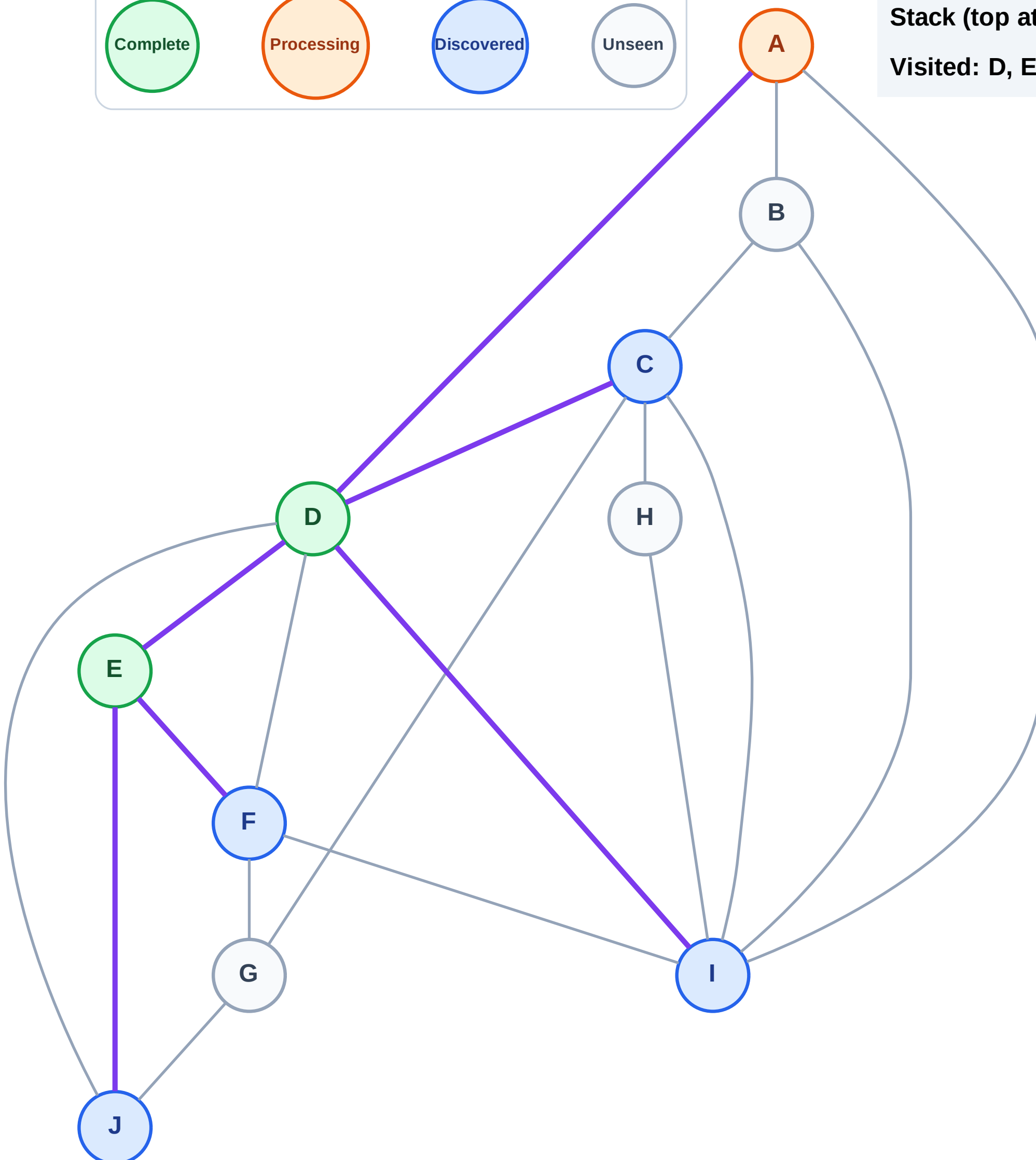
Step 6: Process node A

Legend



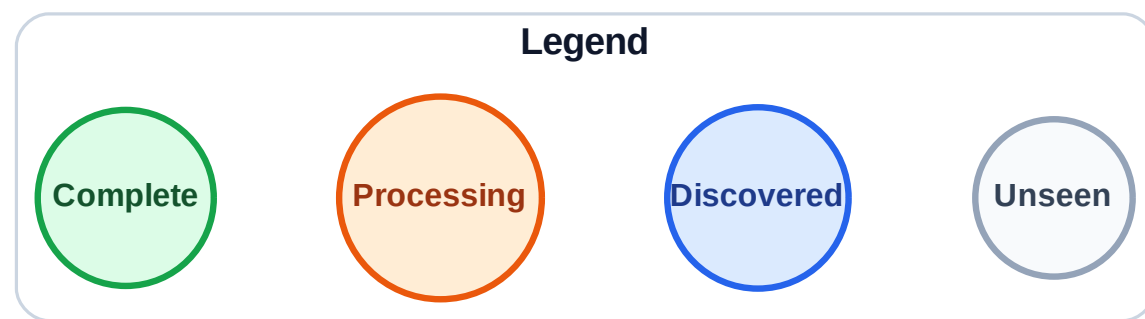
Stack (top at right): J | F | I | C

Visited: D, E

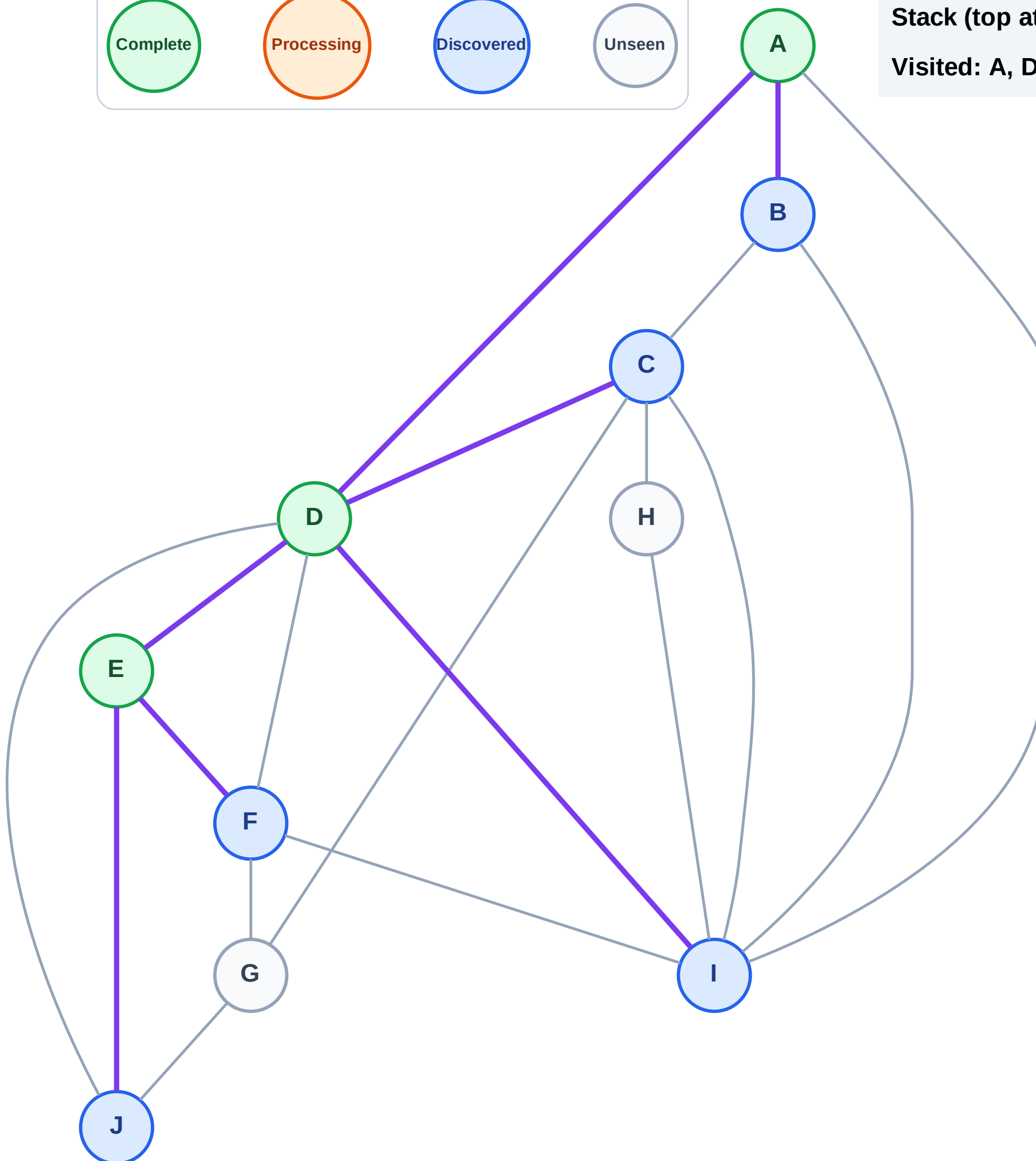


DFS Traversal

Step 7: Discover B from A



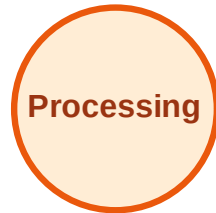
Stack (top at right): J | F | I | C | B
Visited: A, D, E



DFS Traversal

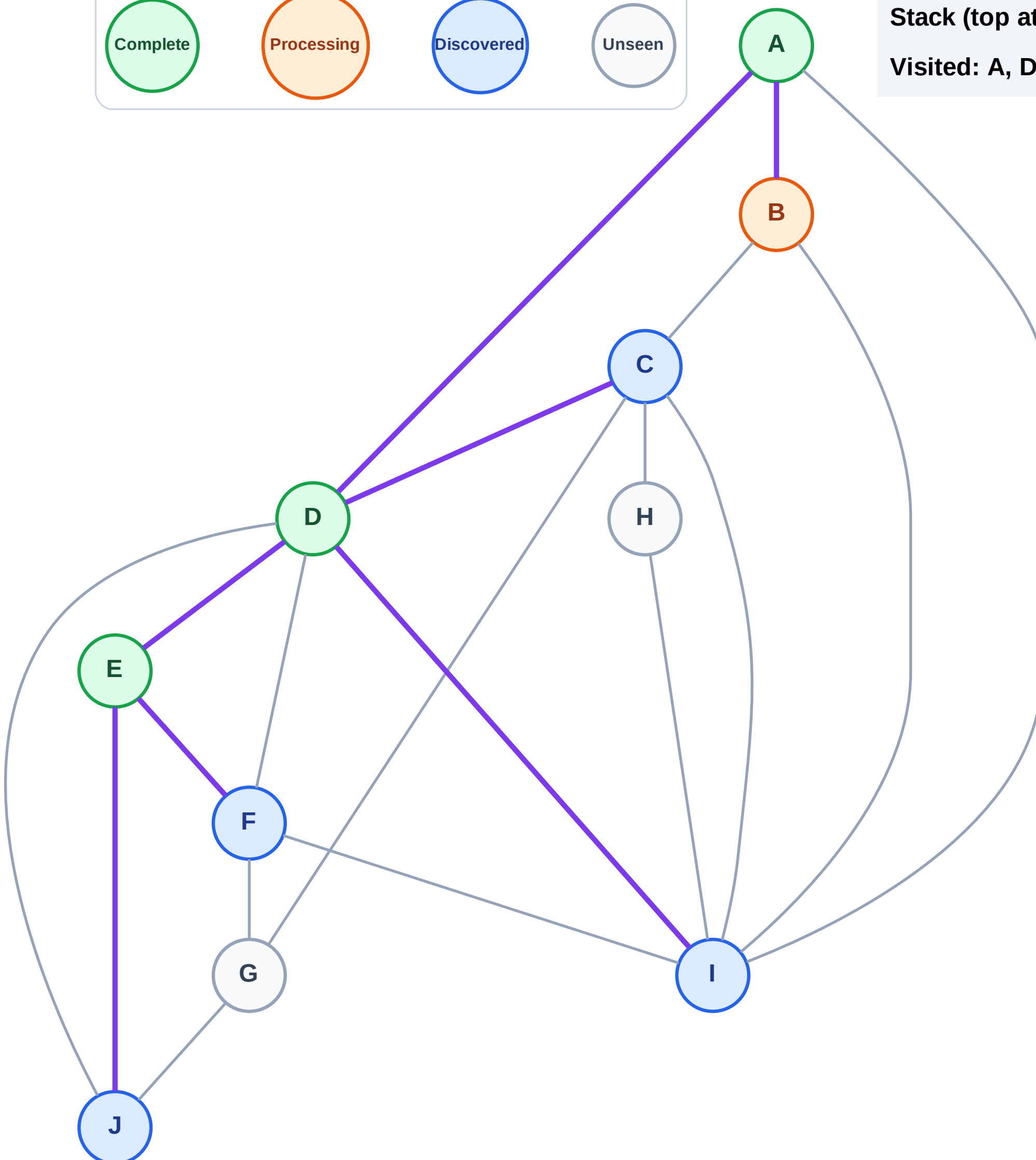
Step 8: Process node B

Legend



Stack (top at right): J | F | I | C

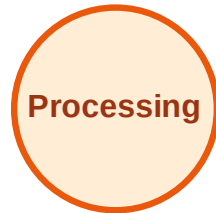
Visited: A, D, E



DFS Traversal

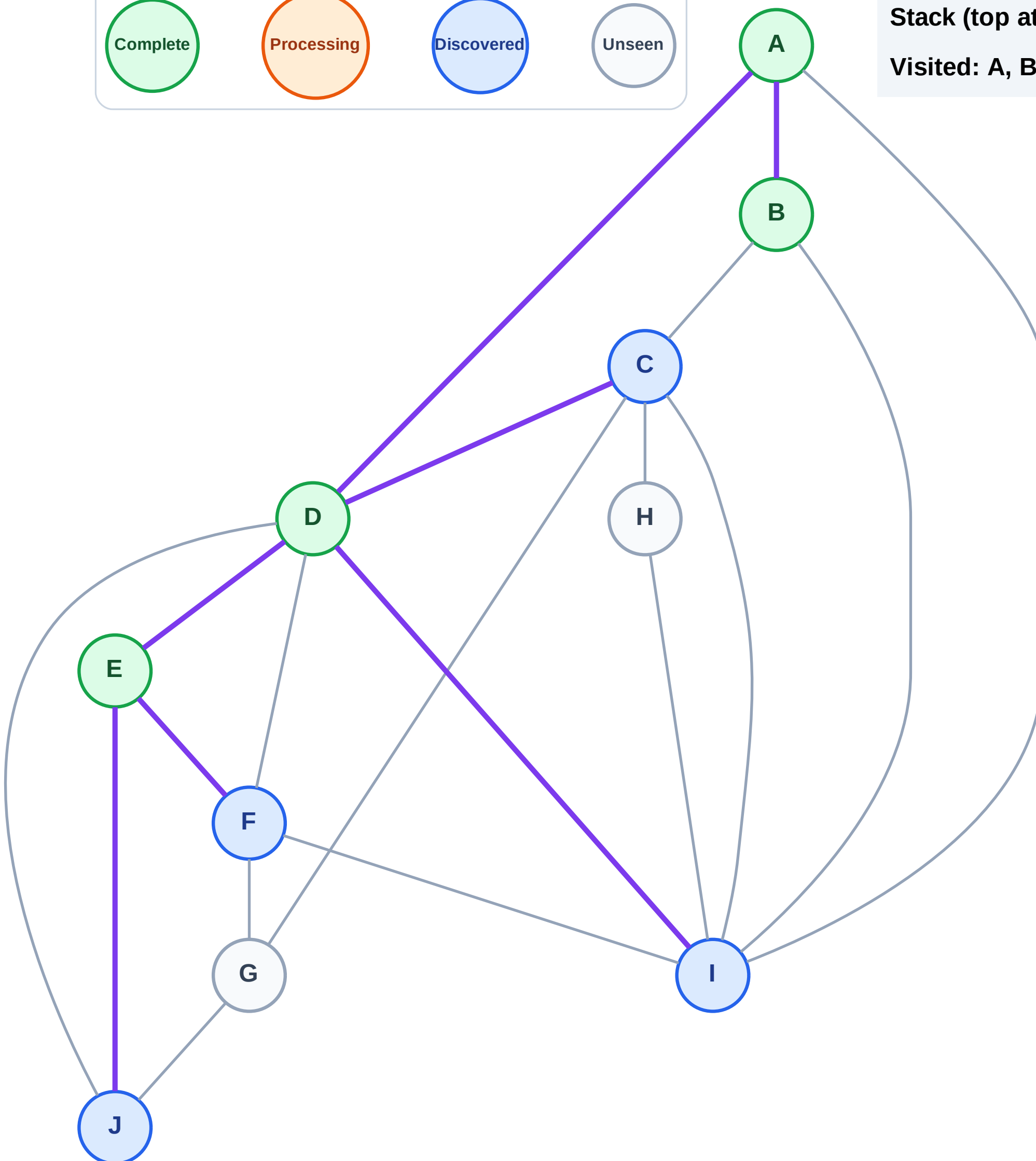
Step 9: No new neighbors from B

Legend



Stack (top at right): J | F | I | C

Visited: A, B, D, E



DFS Traversal

Step 10: Process node C

Legend

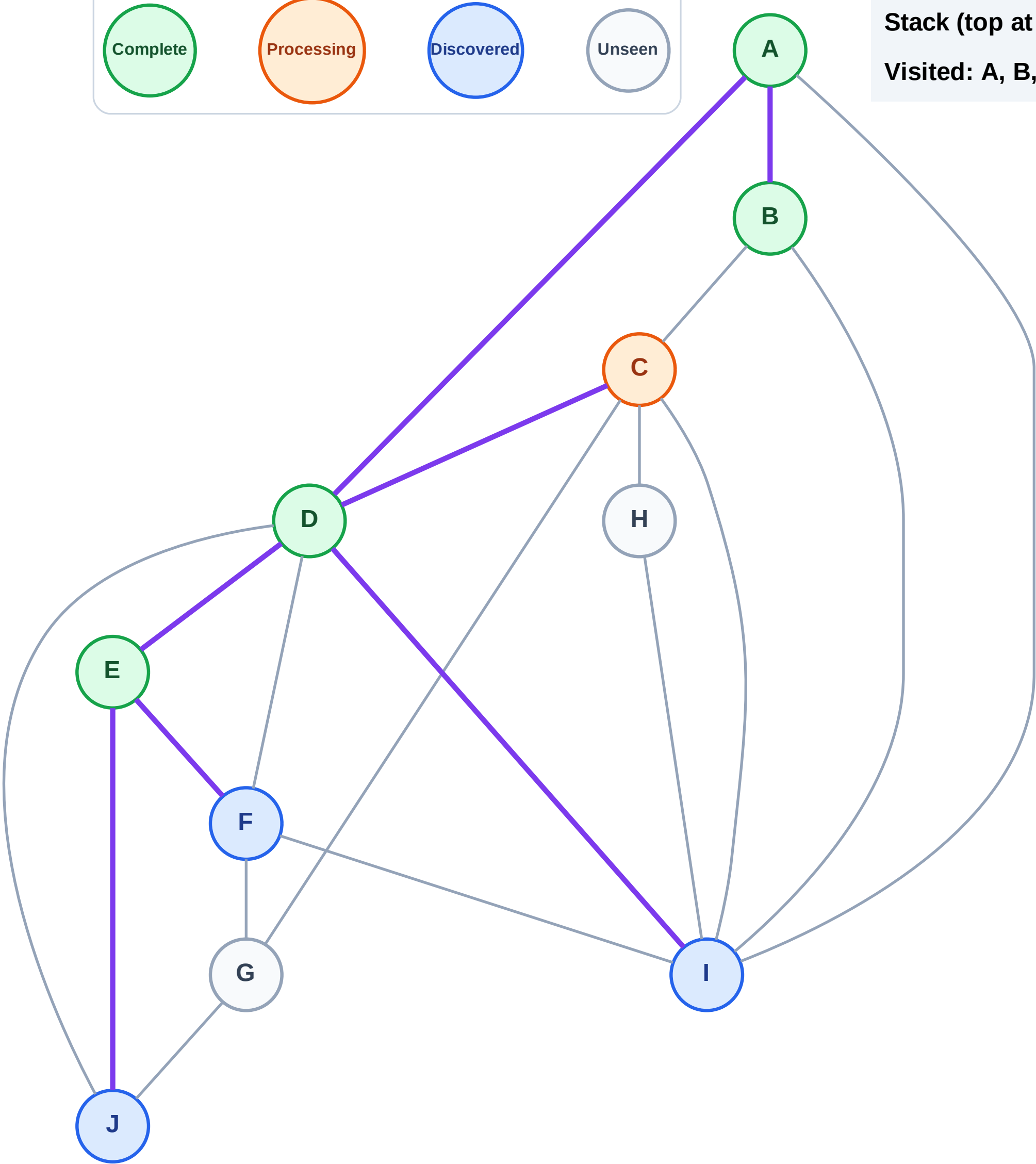
Complete

Processing

Discovered

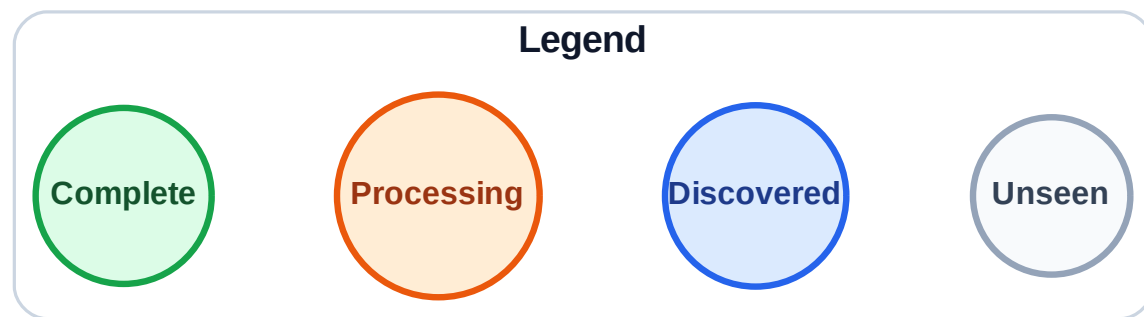
Unseen

Stack (top at right): J | F | I
Visited: A, B, D, E



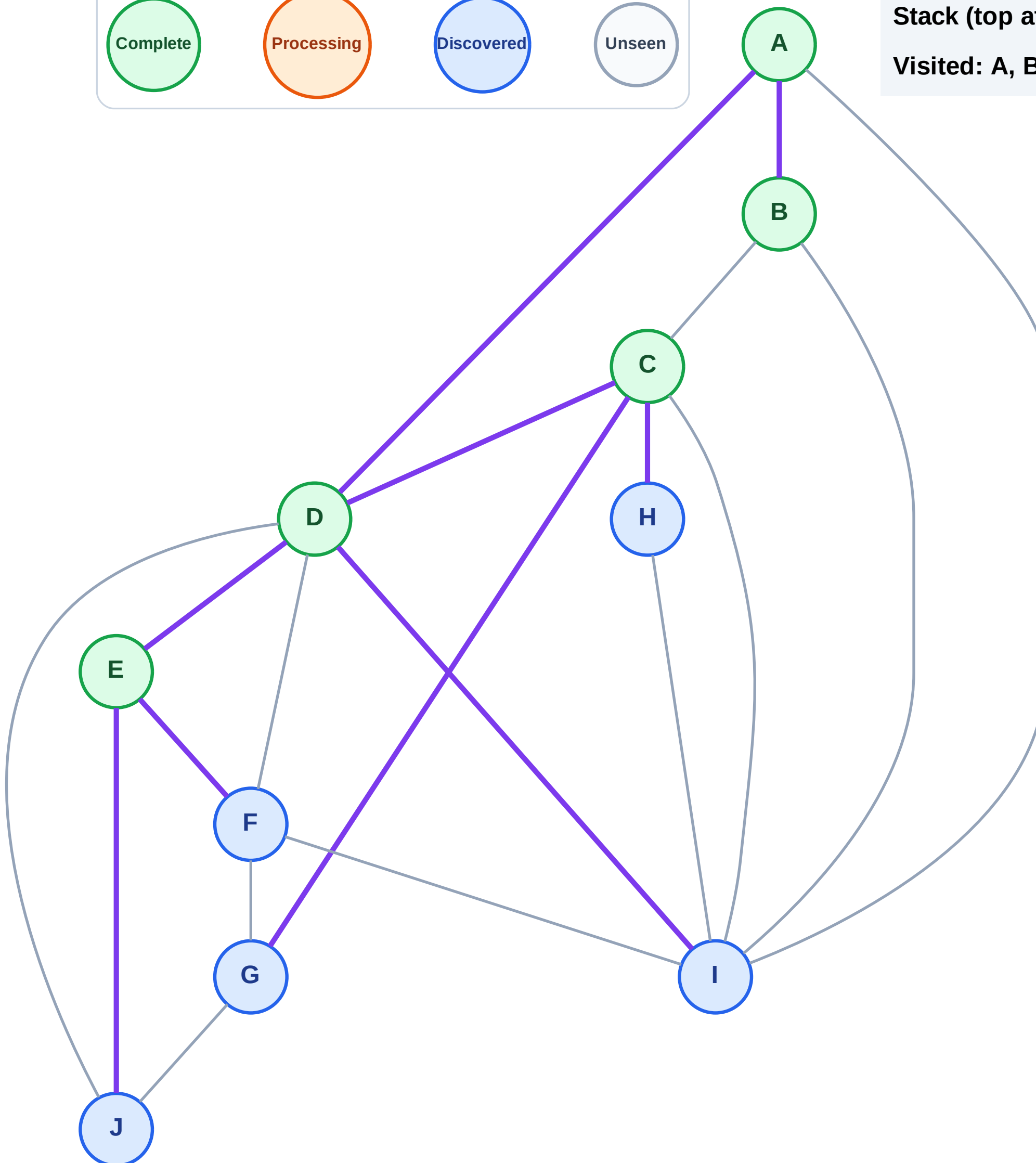
Step 11: Discover H, G from C

Step 11: Discover H, G from C



Stack (top at right): J | F | I | H | G

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node G

Legend

Complete

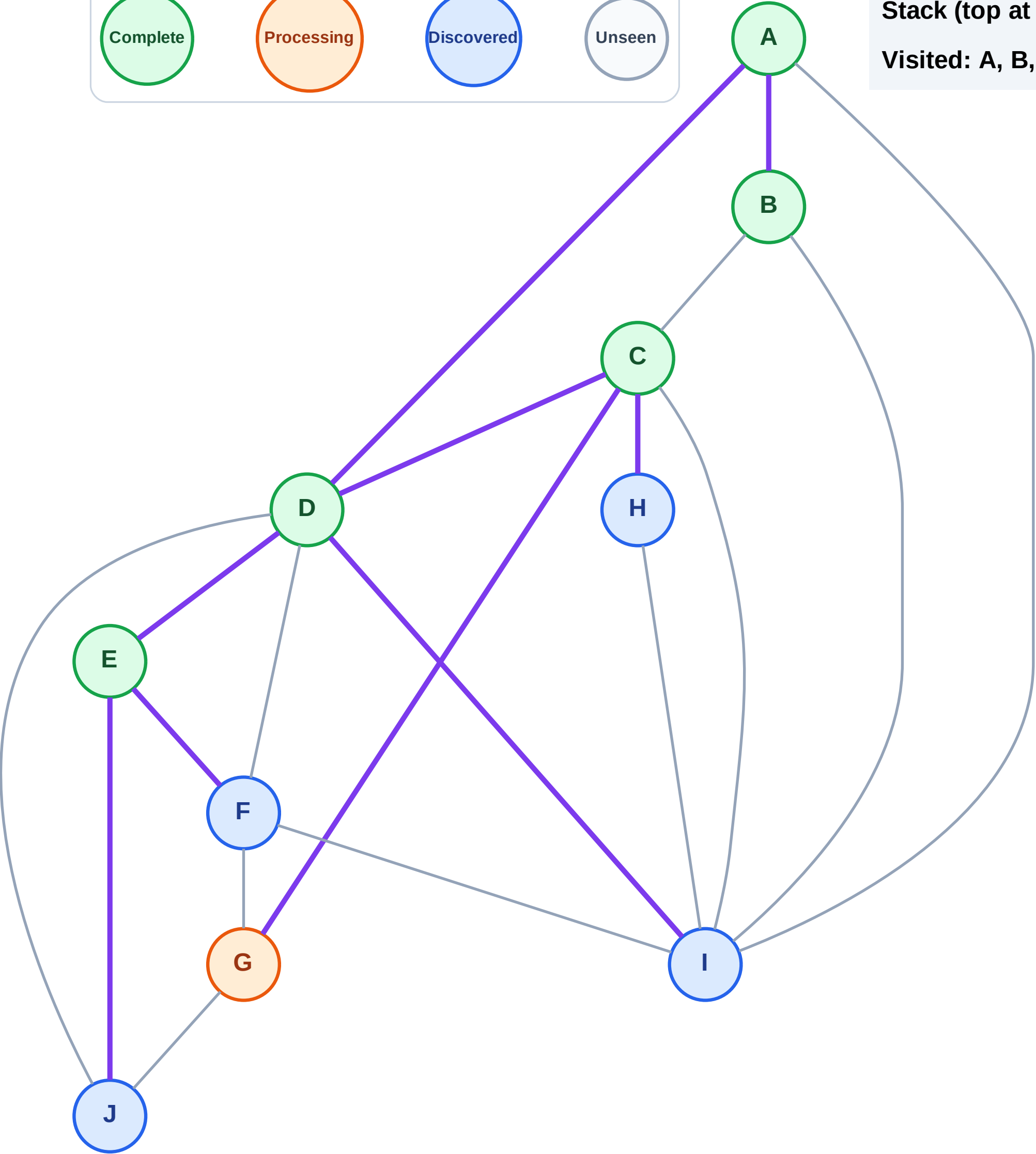
Processing

Discovered

Unseen

Stack (top at right): J | F | I | H

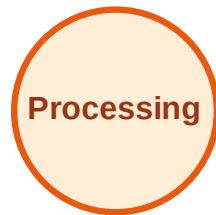
Visited: A, B, C, D, E



DFS Traversal

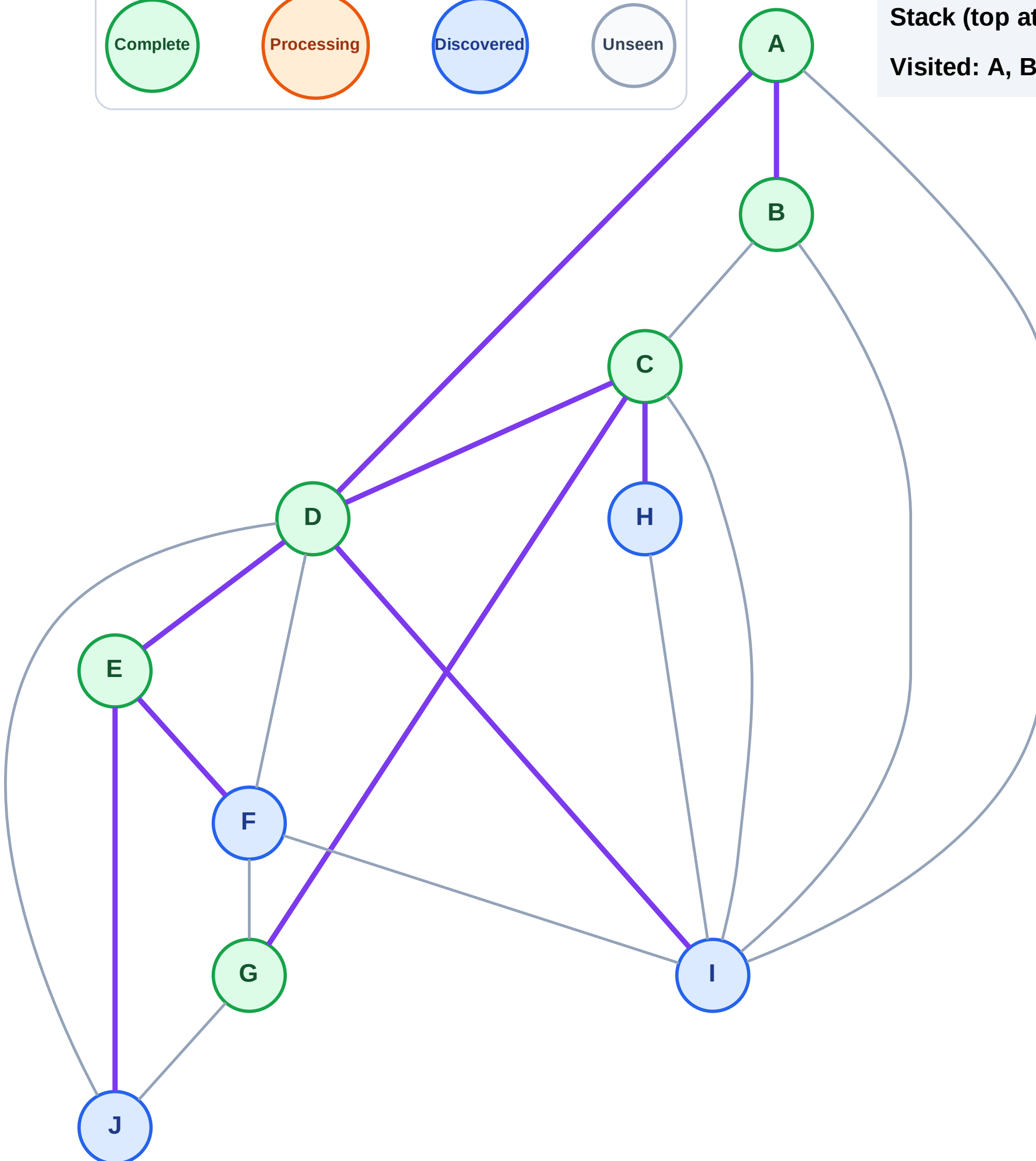
Step 13: No new neighbors from G

Legend



Stack (top at right): J | F | I | H

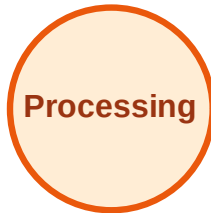
Visited: A, B, C, D, E, G



DFS Traversal

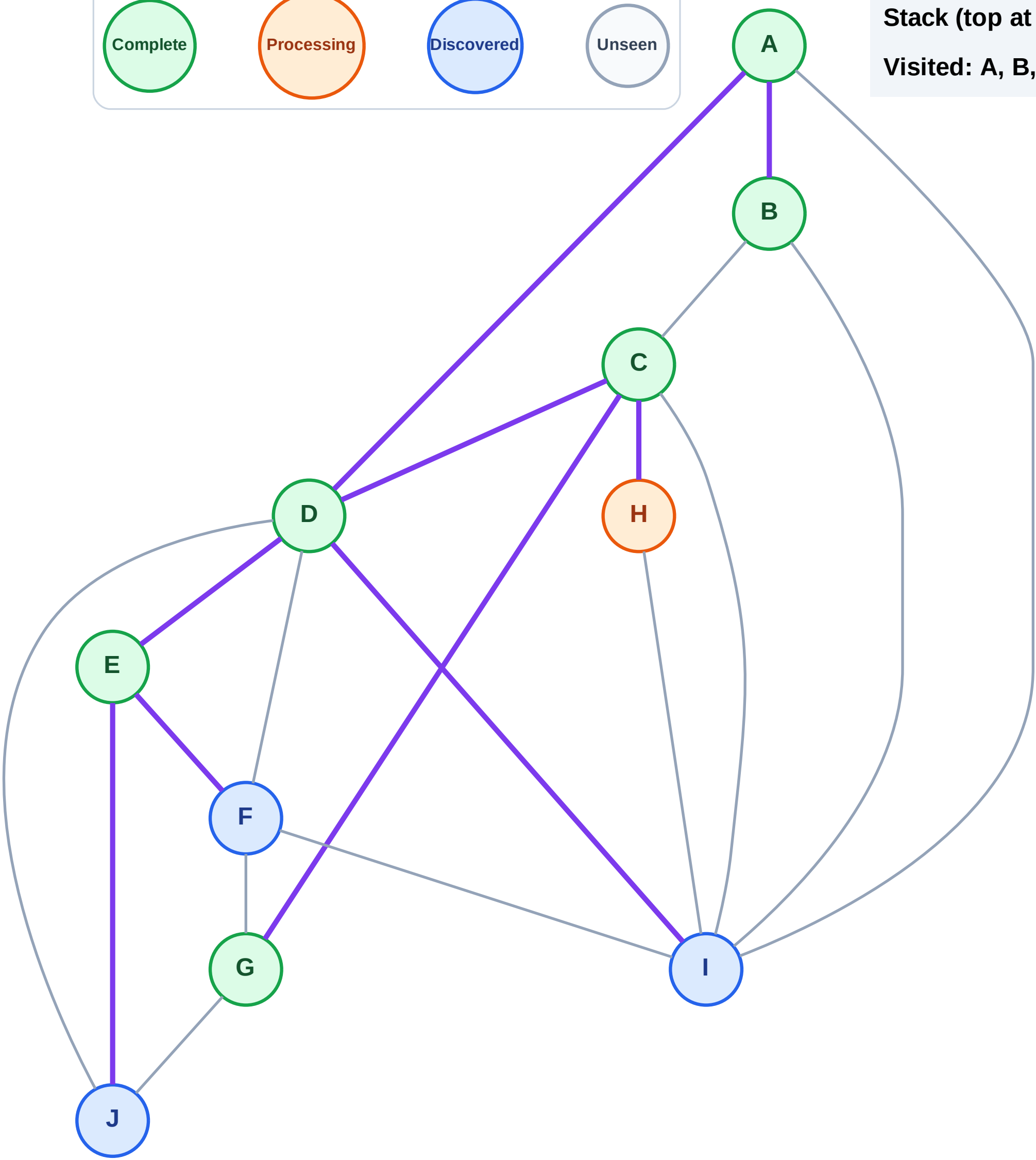
Step 14: Process node H

Legend



Stack (top at right): J | F | I

Visited: A, B, C, D, E, G



DFS Traversal

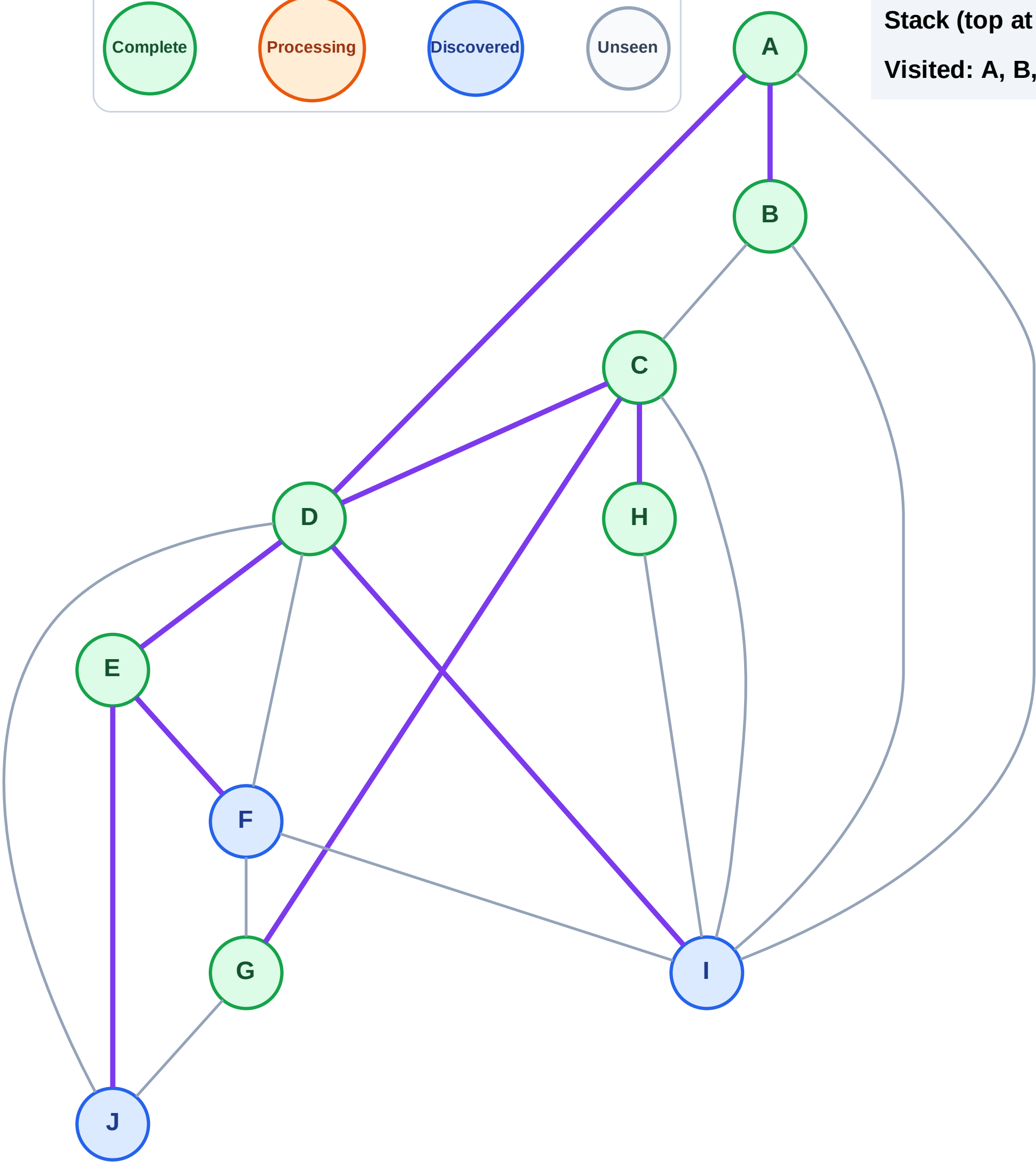
Step 15: No new neighbors from H

Legend



Stack (top at right): J | F | I

Visited: A, B, C, D, E, G, H



DFS Traversal

Step 16: Process node I

Legend

Complete

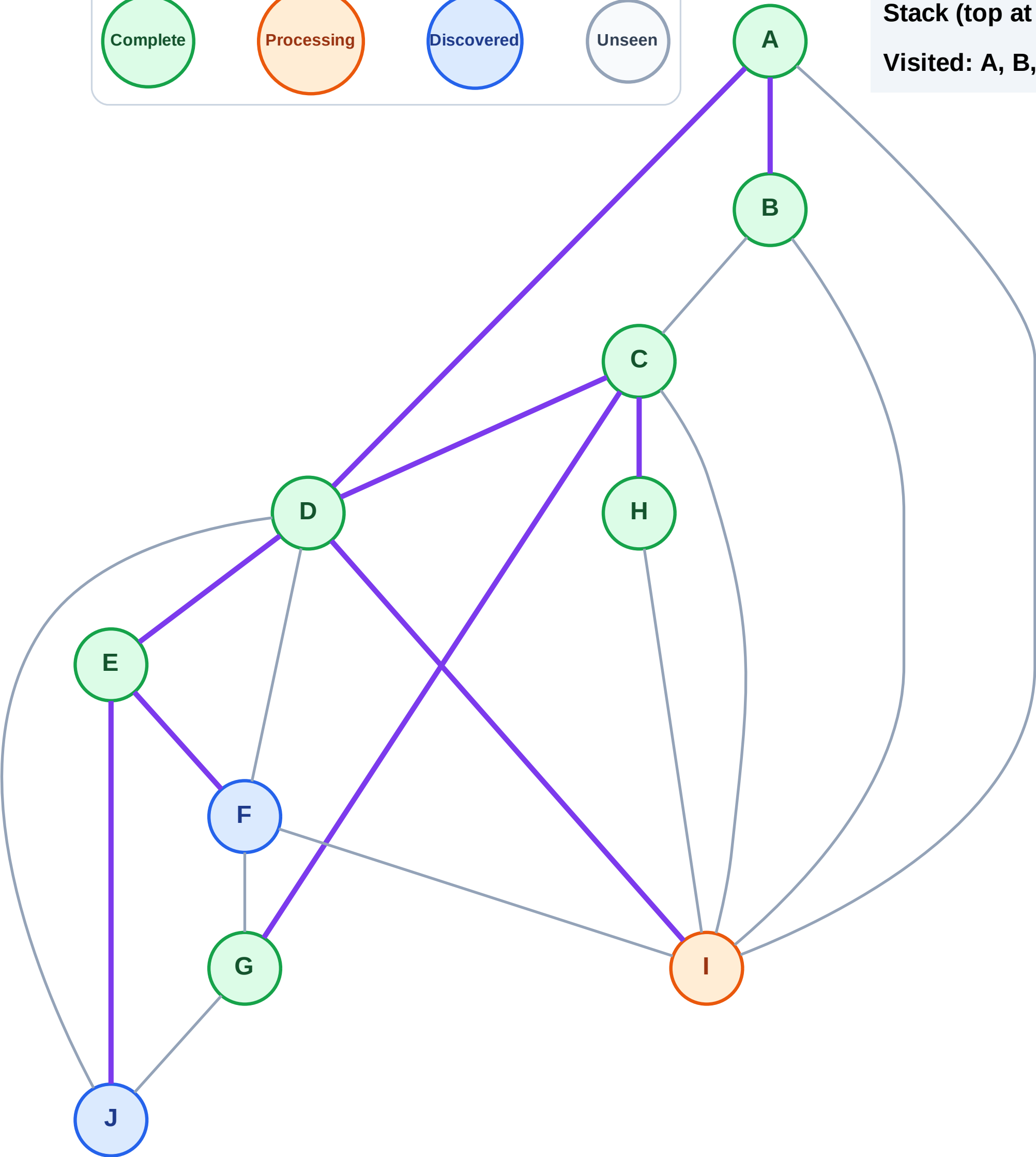
Processing

Discovered

Unseen

Stack (top at right): J | F

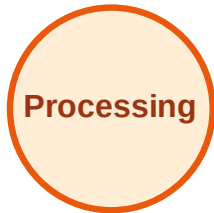
Visited: A, B, C, D, E, G, H



DFS Traversal

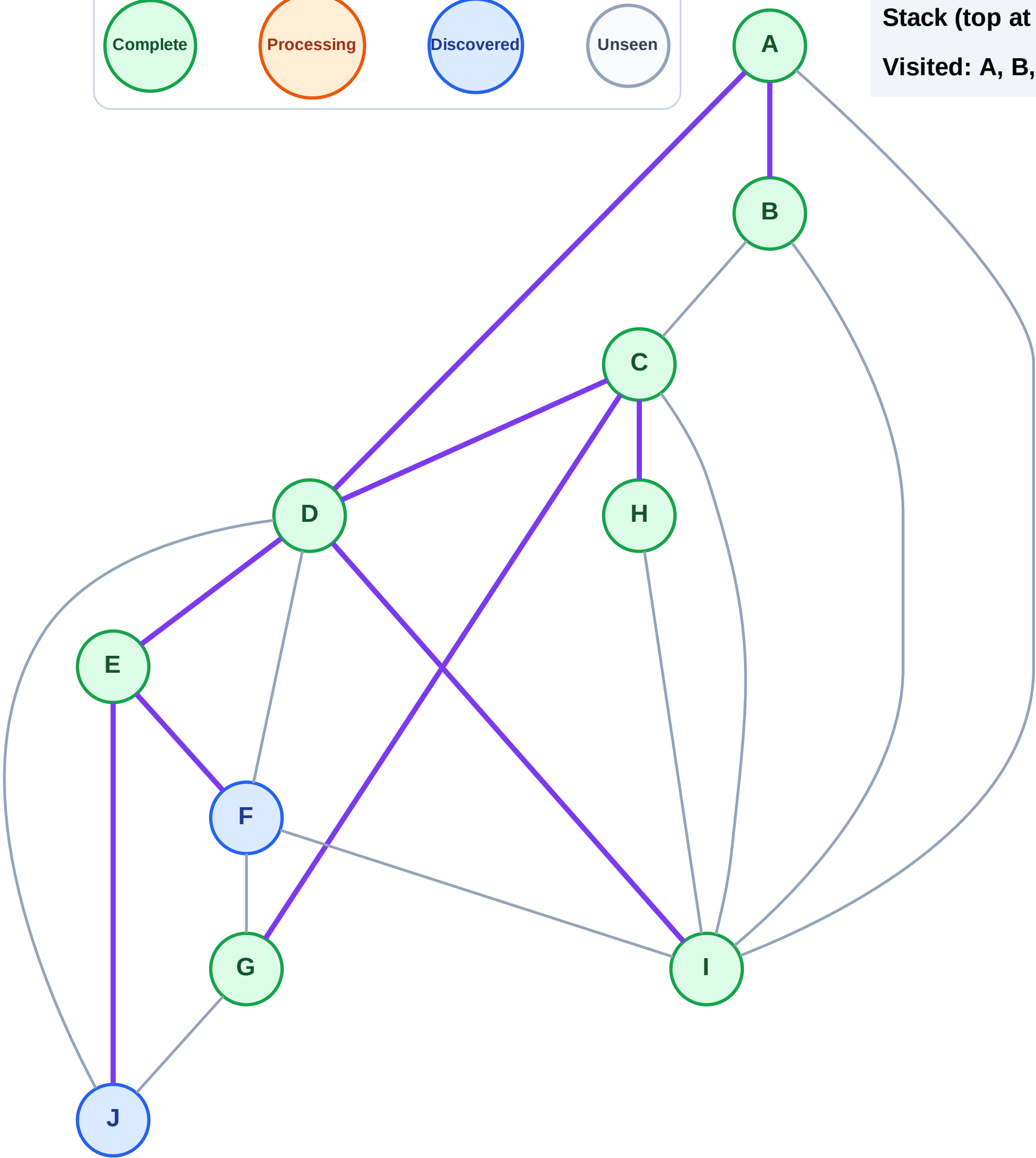
Step 17: No new neighbors from I

Legend



Stack (top at right): J | F

Visited: A, B, C, D, E, G, H, I



DFS Traversal

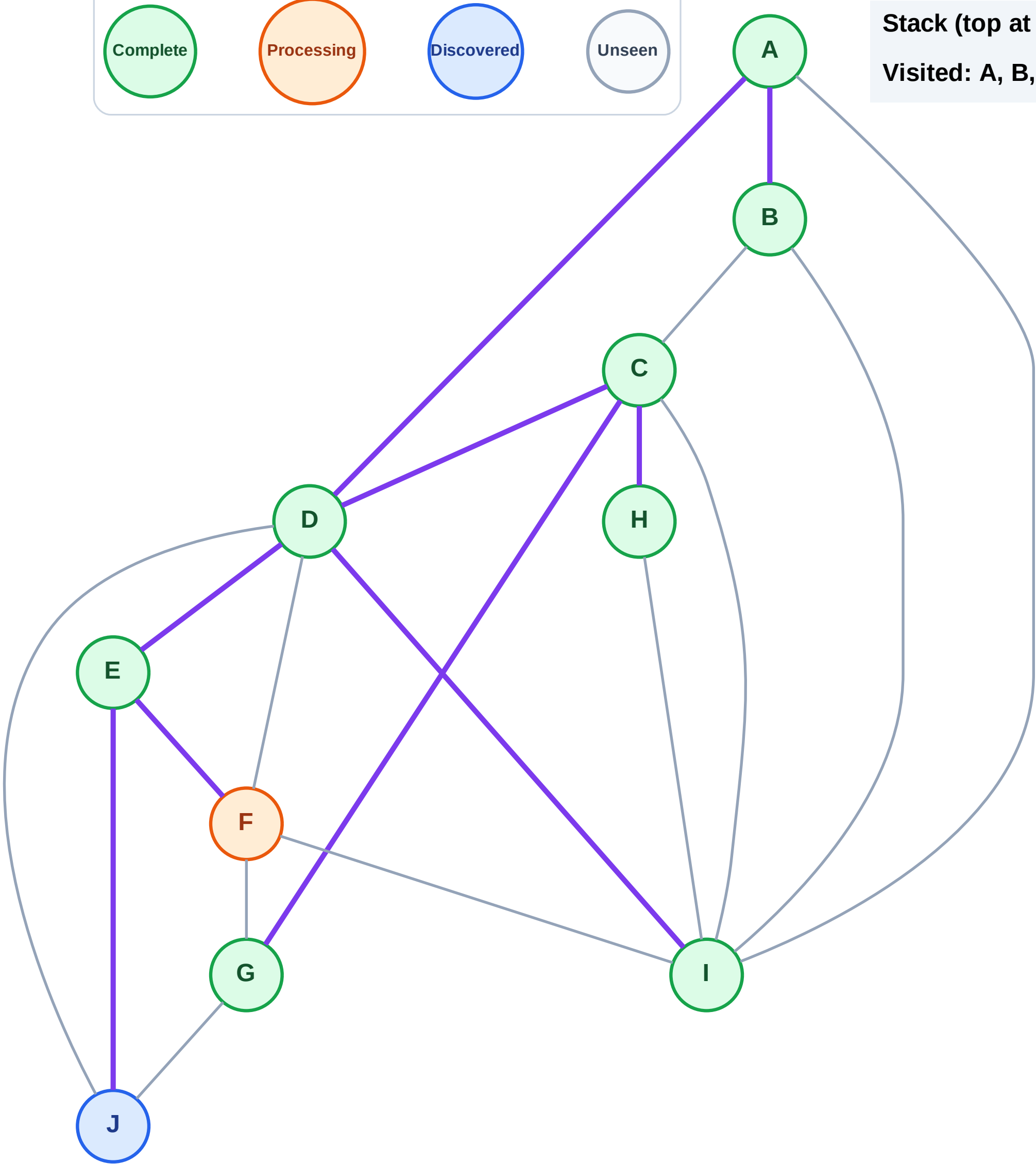
Step 18: Process node F

Legend



Stack (top at right): J

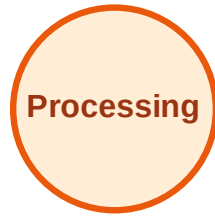
Visited: A, B, C, D, E, G, H, I



DFS Traversal

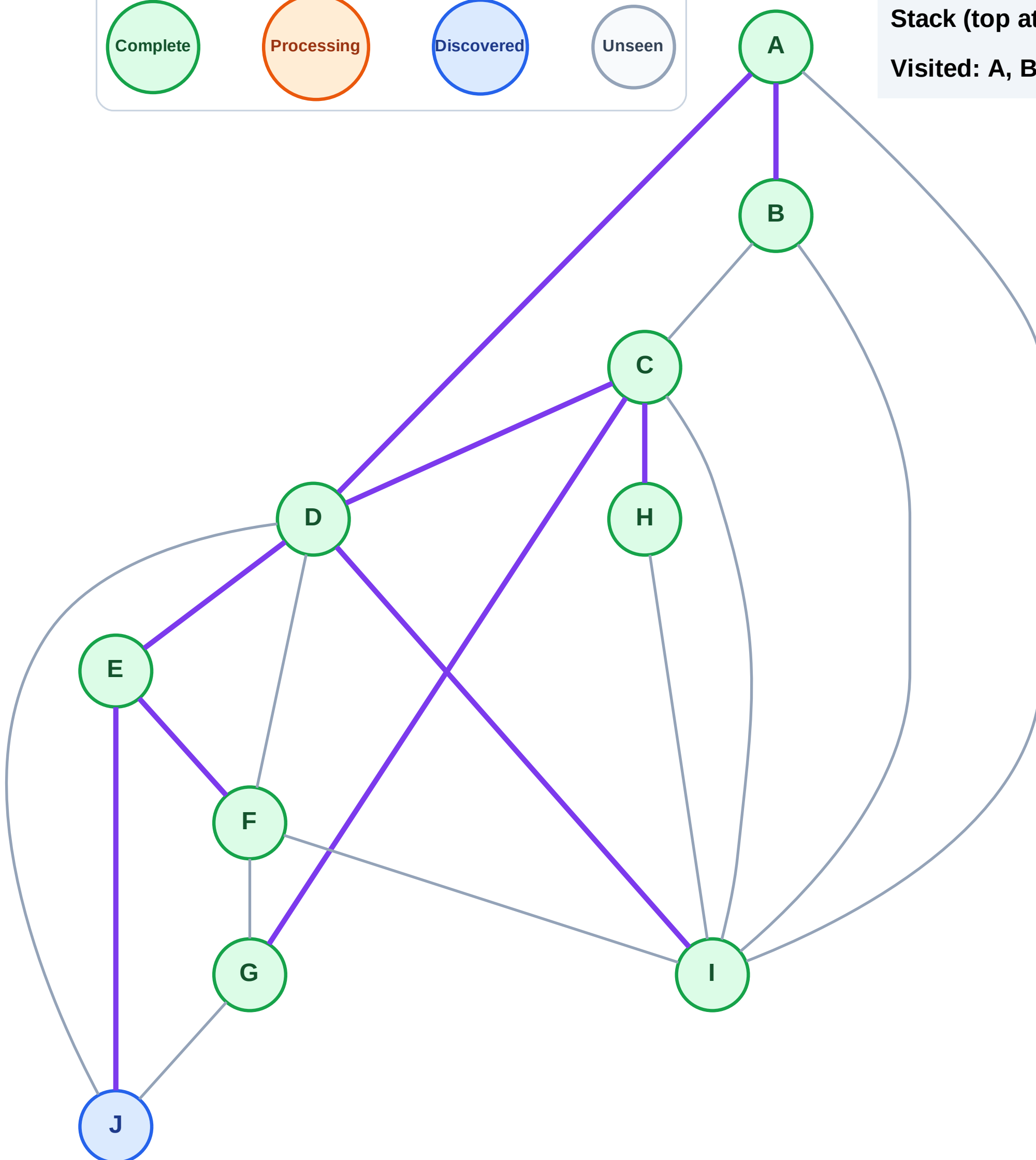
Step 19: No new neighbors from F

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

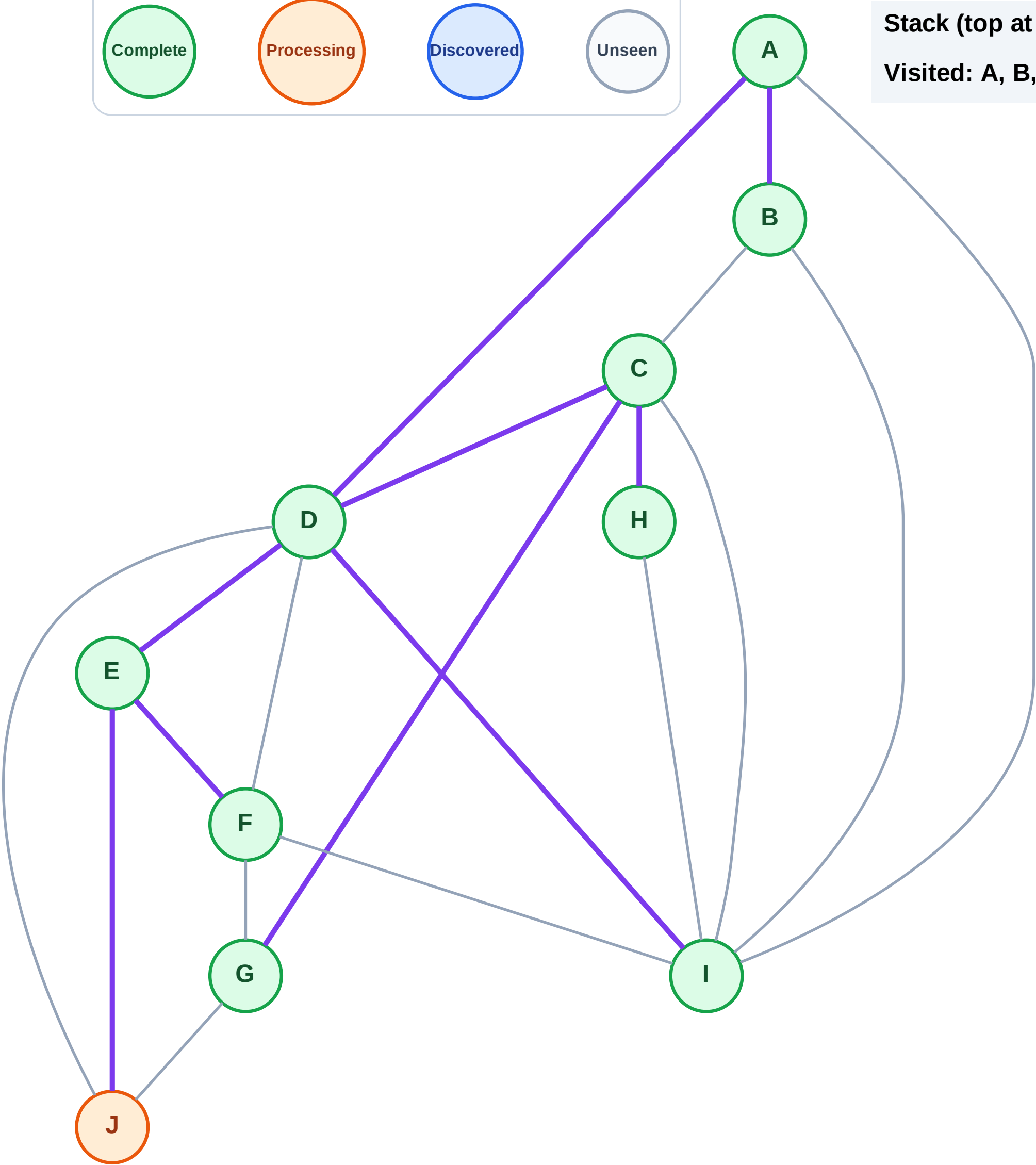
Complete

Processing

Discovered

Unseen

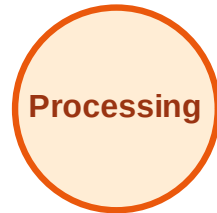
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

